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Observations and Investigations of Two Gravitationally-Lensed Quasars

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Abstract

This thesis presents enhanced Multi Element Remotely Linked Interferometer Network (e-MERLIN) data on two gravitationally-lensed quasars, CLASS B0850+054 and HE 0435-1223. Observations were made with e-MERLIN and data are calibrated with the Astronomical Image Processing System AIPS.

CLASS B0850+054 was observed with e-MERLIN at 5 GHz in search of the central image, and set constraints on the inner mass profile of the lensing galaxy. Although no central image is detected, the detection limit of the observation sets a limit on the flux ratio between the brightest component and the central image. By investigating the Keeton (2003) sample for early-type galaxies and carrying out mass modelling, we have obtained a set of Nuker-law parameters that is able to reproduce the observed source separation and flux ratio. In the next step of modelling, we optimize for the values of convergence κ_b and source separation s while varying the break radius r_b and inner power-law index γ to reproduce the observed data. In addition, the black hole mass is also varied and the sharpness of break α is fitted

for. Results show that either the galaxy has a small core radius and a near-isothermal core, or that the central image is heavily demagnified by the central supermassive black hole (SMBH) and can never be detected.

For the case of the lensed radio-quiet quasar (RQQ) HE 0435-1223, it was observed with e-MERLIN at 1.5 GHz. We suggest that the emission mechanism of this source is star formation around the central SMBH. The map of HE 0435-1223 shows a non-detection of the source, which indicates that the background source size at 1.5 GHz is larger than that of the beam size 408×106 mas^2. This corresponds to an upper limit on the brightness temperature of 1.74×10^3 K at the detection limit of 34.89 μJy beam^{-1}. As a low brightness temperature ($< 10^4$ K) is expected in a star-forming galaxy, we suggest that star formation is powering the radio emission in HE 0435-1223.

Declaration

Except where stated, the work contained in this dissertation is that of the author alone. No portion of the work referred to in this dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institue of learning.

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Preface

First and foremost I would like to express my gratitude to my supervisor Neal Jackson for his untiring support and guidance throughout my MScR, things would be much more difficult without his advice. I would also like to thank Laura Wolz for being my advisor and giving out reminders to inexperienced students like myself. I would like to thank peers and friends at the JBCA for giving me advice and helping me throughout the year.

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This thesis is dedicated to the people fighting for their own rights in Hong Kong. As time gets more and more difficult, I will treasure the value I possess and endure as a Hongkonger.

The Author

The author was born in Hong Kong in February 2000 and attended La Salle College, Hong Kong before obtaining a B.Sc. degree at the University of Hong Kong, graduating in 2022. In September of the same year he started his MScR in Astronomy and Astrophysics at the University of Manchester, Jodrell Bank for Astrophysics Centre.

Chapter 1

Introduction

This thesis investigates the study of gravitationally-lensed quasars. This chapter presents a review on the physics of both gravitational lensing and quasars, and the motivation of this thesis is discussed.

1.1 Historical Background

The history of gravitational lensing can be traced back to the eighteenth century (Newton 1704; Michell 1784), when the corpuscular theory under the classical Newtonian mechanics was proposed. It was predicted that light particles can be deflected by the gravitational attractions of massive bodies. However, the theory for the wave nature of light was more popular in the early nineteenth century, which caused the idea of gravitational lensing to be disregarded. Almost a century later, Einstein (1915) calculated the deflection

of a light ray passing by a massive object under General Relativity, which is a factor of two greater than that under classical Newtonian mechanics. An observation during a solar eclipse was carried out to observe how the apparent position of a star near the limb of the Sun would change, and the result agreed with Einstein's calculations (Dyson et al. 1920).

The extraordinary result raised the attention of astrophysicists to study gravitational lensing. Chwolson (1924) suggested that a ring can be seen if the distant source and the lensing source are perfectly aligned, a phenomenon later called the Einstein ring (e.g. LRG 3-757; Figure 1.1)[1]. Zwicky (1937) predicted that galaxies or galaxy clusters can cause a much stronger lensing effect than a star alone, making more distant sources to be observed. He also proposed that by studying the positions and flux densities of the components in the lens system, it would be possible to study the mass distribution in the lenses. This later became one of the most studied topics in the field.

Quasars, also known as Quasi-Stellar Objects (QSOs), are extremely bright and distant point-sources (Burke, Graham-Smith & Wilkinson 2019). A quasar resides at the centre of a host galaxy and outshines it due to their extreme brightness. In quasar observations, characteristic broad emission lines redshifted to the optical wavelengths are observed (Berk et al. 2001). Schmidt (1963) made the discovery and it boosted the study of gravitational lensing enormously. It was thought that quasars, together with a nearer massive source, are suitable candidates for studying gravitational lensing as they

[1] `https://apod.nasa.gov/apod/ap111221.html`

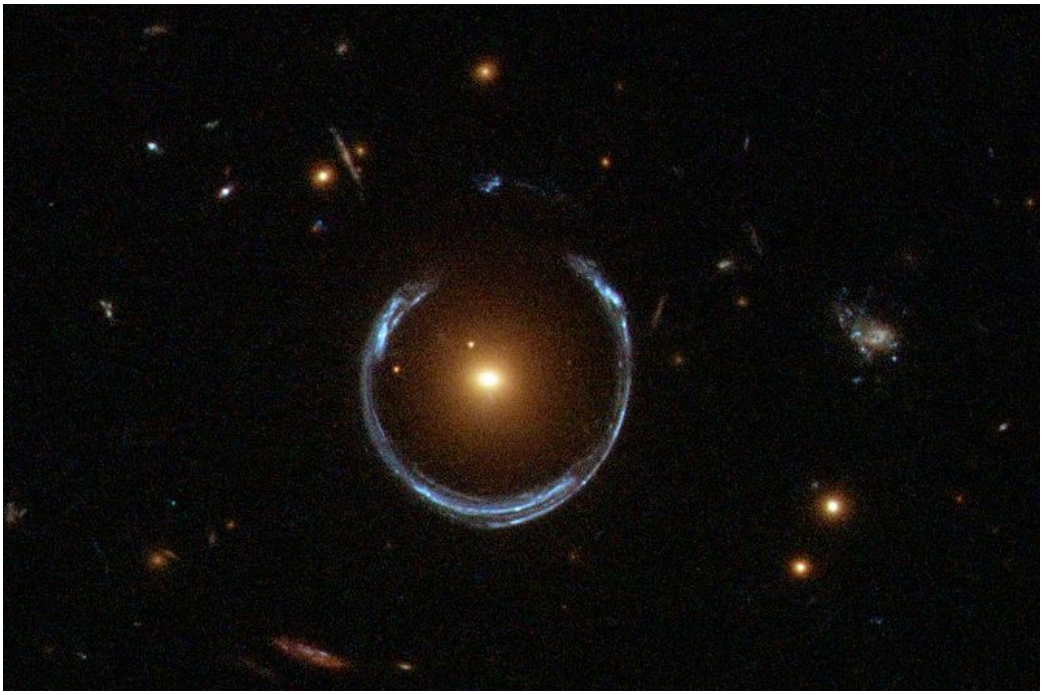


Figure 1.1: An Einstein ring in the gravitational lens system LRG 3-757.

are easy to be recognised. In addition, the variability of quasars provides a suitable condition for determining the H_0 constant. In fact, the first identified gravitational lens contains a quasar QSO 0957+561 (Walsh, Carswell & Weymann 1979). The two components in the lens system were confirmed to be the same source for their similar flux ratios between optical and radio wavelengths and their same redshifts.

1.2 Gravitational Lensing

This section is a review on the basics of gravitational lensing and its relevant theories based on Schneider, Ehlers & Falco (1992), Kochanek (2004), and Dodelson (2017). The motivation of studying gravitational lensing is also discussed.

1.2.1 Deflection Angle

We can begin the discussion by considering the deflection angle equation, which was derived by Einstein (1915) using General Relativity. Suppose a light ray passes by a point mass M and assuming small deflection angles and weak gravitational fields, the deflection angle $\hat{\alpha}$ of the light ray is given by

$$\hat{\alpha}(\xi) = \frac{4GM}{c^2\xi}, \tag{1.1}$$

where G is the gravitational constant, c is the speed of light in vacuum, and ξ is the impact parameter of the light ray. Generally, if the lens is not infinitely small but is an extended mass distribution, it is necessary to consider the contribution of each individual deflection by different components of the lens. Therefore, the deflection angle of a light ray passing by an extended mass distribution is given by

$$\hat{\alpha}(\xi) = \frac{4G}{c^2} \iint \frac{\Sigma(\xi')(\xi - \xi')}{|\xi - \xi'|^2} \, d^2\xi', \tag{1.2}$$

where $\Sigma(\xi')$ is the surface mass distribution

$$\Sigma(\xi') = \int \rho(\xi', z) \, dz, \tag{1.3}$$

where ρ is the density of the lens, ξ' is the distance from the centre of the lens, and z is the distance along the optical axis.

1.2.2 Lens Equation

To construct the lens equation, we can first consider the geometry of a lens system (Figure 1.2). The system can be divided into the source plane, the lens plane and the observer. The source plane is where the true source position S is, at angular distance β from the optical axis as seen by the observer O. The lens plane consists of the lensing mass distribution (e.g. a galaxy) and the optical axis passes through the centre of the lens. Light rays from the source

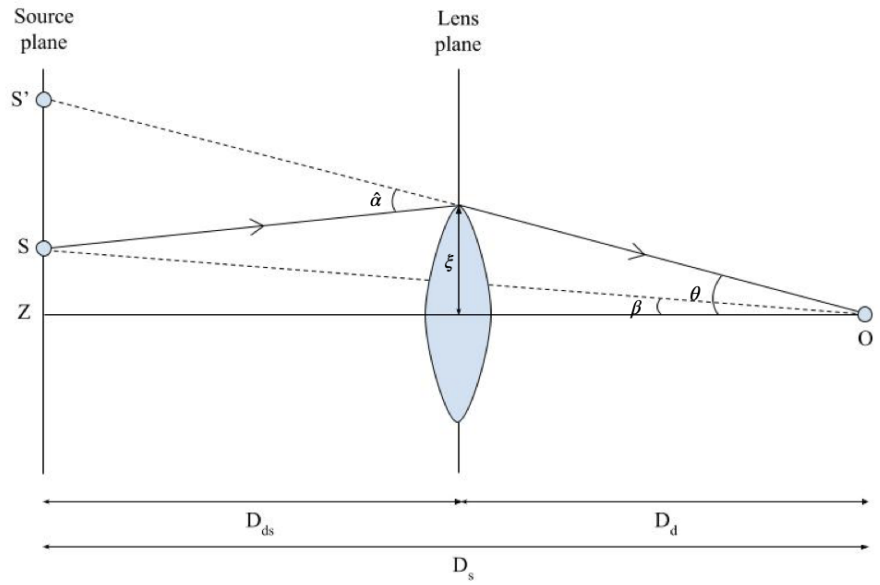


Figure 1.2: Lens geometry diagram showing the deflection of light from the source (S) to the observer (O) due to a mass distribution in the lens plane. The light ray experiences a deflection of angle $\hat{\alpha}$ at impact parameter ξ away from the lens centre. Projecting the light ray backwards from the lens plane to the source plane gives the apparent source position (S'). The true and apparent source positions are at angles β and θ away from the optic axis respectively. The distances between the source plane and the lens plane, the lens plane and the observer, and the source plane and the observer are D_{ds}, D_d and D_s respectively.

experience a deflection at the lens plane by angle $\hat{\alpha}$ and reach the observer. By projecting the deflected light rays back to the source plane, the apparent source position S', that is at angular distance θ away from the optical axis as seen by the observer can be obtained. The distances between the source plane and the lens plane, the lens plane and the observer, and the source plane and the observer are D_{ds}, D_d and D_s respectively. At the source plane, the relation between the true and apparent source positions is given by

$$ZS' - ZS = SS', \tag{1.4}$$

where Z is the origin of the optical axis. Using simple trigonometry, the lens equation can be obtained

$$\theta - \beta = \hat{\alpha}\frac{D_{ds}}{D_s}. \tag{1.5}$$

1.2.3 Angular Diameter Distance

In Figure 1.2, $D_{ds}+D_d = D_s$ holds in Euclidean coordinates when the distance between the source and the observer remains relatively small. However, when observing a distant source ($z > 1$), the sum of the distances D_d and D_{ds} is not equivalent to D_s in the non-Euclidean regime. Therefore, angular diameter distances are introduced to ensure that the lens equation (Equation 1.5) holds true even in cosmological distances. The comoving radial distance D_{ij}

between two redshifts i and j is given by Kochanek (2004)

$$D_{ij} = \frac{r_H}{|\Omega_k|^{1/2}} \mathrm{sinn}\Big[\int_i^j \frac{|\Omega_k|^{1/2}\, dz}{[(1+z)^2(1+\Omega_M z) - z(2+z)\Omega_\Lambda]^{1/2}}\Big], \tag{1.6}$$

where Ω_M, Ω_Λ, and $\Omega_k = 1 - \Omega_M - \Omega_\Lambda$ are the present day matter density, the cosmological constant, and the 'curvature' density respectively, and $r_H = c/H_0$ is the Hubble radius. The $\mathrm{sinn}(x)$ function depends on whether the model of the universe is open ($\Omega_k > 0$), flat ($\Omega_k = 0$) or closed ($\Omega_k < 0$) (Carroll, Press & Turner 1992), such that

$$\mathrm{sinn}(x) = \begin{cases} \sinh(x) & \text{for } \Omega_k > 0, \\ x & \text{for } \Omega_k = 0, \\ \sin(x) & \text{for } \Omega_k < 0. \end{cases} \tag{1.7}$$

Ultimately, the angular diameter distance (D_{ij}^{ang}) is given by

$$D_{ij}^{ang} = \frac{D_{ij}}{1+z}. \tag{1.8}$$

1.2.4 Magnification

Gravitational lensing can alter the effective area of a background source by increasing its angular size. Since the surface brightness of a source is conserved, gravitational lensing therefore increases the observed flux density of a source, allowing us to detect a more distant object. The magnification μ

of a lensed image is defined as the ratio between the image and the source flux densities, and can be simplified to the ratio between the solid angle subtended by the image $d\omega$ and the source $d\omega'$

$$\mu = \frac{d\omega}{d\omega'}. \tag{1.9}$$

An alternative approach is to consider the Jacobian matrix in the image ($\delta\theta$) and the source planes ($\delta\beta$). The Jacobian matrix (A_{ij}) is given by

$$A_{ij} = \frac{\partial\beta_i}{\partial\theta_j}, \tag{1.10}$$

where i and j represent the dimensions of each plane. By taking the lens equation (Equation 1.5) and differentiate w.r.t. θ, it can be written as

$$\delta_{ij} - A_{ij} = \nabla_\theta\hat{\alpha}, \tag{1.11}$$

where δ is the Kronecker delta function (value of 1 if $i = j$ and 0 otherwise). The deflection angle $\hat{\alpha}$ can be written as

$$\hat{\alpha} = \nabla\psi, \tag{1.12}$$

where ψ is the gravitational potential of the lens projected onto the 2-

dimensional lens plane. Therefore, Equation 1.11 can be re-written as

$$A_{ij} = \delta_{ij} - \nabla^2\hat{\psi}. \tag{1.13}$$

Defining the shears γ_1, γ_2 and convergence κ

$$\gamma_1 = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} - \frac{\partial^2\psi}{\partial\theta_2^2}\right), \tag{1.14}$$

$$\gamma_2 = \frac{\partial^2\psi}{\partial\theta_1\partial\theta_2} = \frac{\partial^2\psi}{\partial\theta_2\partial\theta_1}, \tag{1.15}$$

and

$$\kappa = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} + \frac{\partial^2\psi}{\partial\theta_2^2}\right), \tag{1.16}$$

we can solve the Jacobian matrix A and the magnification μ (Equation 1.13)

$$A = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}, \tag{1.17}$$

$$\mu = \frac{1}{\det|A|} = [(1-\kappa)^2 - \gamma^2], \tag{1.18}$$

where $\gamma = \gamma_1^2 + \gamma_2^2$. Convergence and shears describe the change in area and shape of the image in different manners. Convergence κ depends on the surface density of the lens where light rays pass through, and changes the cross-sectional area of ray bundles of light from the source in spherically symmetric manner. Shears γ, on the other hand, depend on the matter

outside of where the light rays pass through, and represent the distortions of the image after passing through the lens.

1.2.5 Mass Modelling

The Schwarzschild lens is a singularity of mass M to model the lensing galaxy in the simplest way (Refsdal 1964). However, extended mass profiles are usually used to model galaxies that are responsible for strong lensing in a more practical manner. Such a mass profile has a surface density that changes with radius

$$\rho(r) \propto \frac{1}{r^{\gamma}} \frac{1}{\left(a^{\alpha} + r^{\alpha}\right)^{(\beta-\gamma)/\alpha}}, \tag{1.19}$$

where $r \simeq a$ represents the break radius of the profile, and β, γ, and α represent the outer power-law index, inner power-law index, and the sharpeness of break respectively. A central density cusp with $\rho \propto r^{-\gamma}$ is also included. For instance, the singular isothermal sphere (SIS) is a mass profile without a core radius (Hinshaw & Krauss 1987; Kochanek 2004). The 3-dimensional density distribution $\rho(r)$ of an SIS is given by

$$\rho(r) = \frac{\sigma^2}{2\pi G} \frac{1}{r^2}, \tag{1.20}$$

where σ is the velocity dispersion of the galaxy along the line of sight and r is the radial distance from the centre of the galaxy. By Equation 1.3, the

2-dimensional surface mass distribution of an SIS is given by

$$\Sigma(r) = \frac{\sigma^2}{2G}\frac{1}{r}. \tag{1.21}$$

Other common profiles include the Hernquist model ($\gamma = 1$, $\beta = 4$, $\alpha = 1$), the Jaffe model ($\gamma = 2$, $\beta = 4$, $\alpha = 1$), the NFW model ($\gamma = 1$, $\beta = 3$, $\alpha = 1$), and the Moore model ($\gamma = 3/2$, $\beta = 3$, $\alpha = 1$; Hernquist 1990; Jaffe 1983; Navarro, Frenk & White 1997; Moore et al. 1998). Generally, the assumption of thin lens approximation, where the thickness of the lens is much smaller than the distance that the light ray has travelled, is made. Blandford & Narayan (1986) have studied multiple deflection screens but they are usually used in numerical studies only, due to the lack of constraints in observational studies for such models.

Wallington & Narayan (1993) suggested that the central region of a mass distribution (at radius of tens of parsecs) is usually dominated by a central SMBH and a central stellar cusp. A broken power-law with a break radius separating the inner and outer regions of a galaxy can be used in a model to manifest the mass distribution of a galaxy containing a central SMBH or a central stellar cusp (e.g. Keeton 2003). Ferrarese & Merritt (2000) suggested that by studying the gas disc that orbits around the SMBH and measuring its velocity dispersion σ (in units of km s^{-1}) along the line of sight, one can find the black hole mass M_{BH} by using a strong correlation. A popular relation

between M_{BH} and σ by Gebhardt et al. (2000) is given by

$$\frac{M_{BH}}{M_{\odot}} = 0.282\sigma^{3.75}, \tag{1.22}$$

where $M_{\odot}$ is the mass of the Sun. On the other hand, a surface density distribution of the following form can be used for mass modelling

$$\kappa(r) \propto \frac{1}{r^{\gamma-1}} \frac{1}{(a^{\alpha}+r^{\alpha})^{(\beta-\gamma)/\alpha}}. \tag{1.23}$$

Such a surface density model is called the Nuker law (Byun et al. 1996; Faber et al. 1996). Expressing the Nuker law in terms of surface brightness distribution (Keeton 2003) gives

$$I(r) = 2^{(\beta-\gamma)/\alpha} I_b \left(\frac{r}{r_b}\right)^{-\gamma} \left[1+\left(\frac{r}{r_b}\right)^{\alpha}\right]^{(\gamma-\beta)/\alpha}, \tag{1.24}$$

where r_b is the break radius (point of change in inner and outer power-law indices), and I_b is the surface brightness at the break radius. The Nuker law will be further discussed in Section 3.5.2.

Circular lenses, which require fewer constraints on mass models, are popular in the study of gravitational lensing as their lens equations can be solved by using simple graphical interpretations. In the case of searching for core images in lenses, ellipticity is often ignored as it only has a small effect on the core image magnification (Keeton 2003). Given the lens model and the source position, graphical solutions tell us how far images are formed from

the lens centre, as well as their magnifications. By plotting the lens equation (Equation 1.5) and the bend angle curve on the same graph, the positions of the images can be directly read off at the intersections of the two curves (Figure 1.3).

1.2.6 Motivation

Theoretical predictions of gravitational lensing suggest that both two- or four-image lens systems should have an additional central image in the vicinity of the lensing galaxy if the central mass distribution is less steep than $\rho \propto r^{-2}$ (Burke 1981). By detecting the central image and determining its brightness, tighter constraints can be set on the mass distribution of the lensing galaxy. However, the detection of such an image is extremely rare. Keeton (2003) stated that since central images are very sensitive to the central mass distributions of lens galaxies, their demagnifications due to a central SMBH can range from 10^{-3} to 10^{-5}. In addition, the lensing galaxy obscures the central image, making it even more difficult to be detected. On the other hand, the detection of a central image is also affected by the configuration of a lens system. In particular, a two-image lens system with the source located near the Einstein radius provides the best scenario for central image detection (Quinn et al. 2016). Figure 1.3 shows the bend angle diagram of such a scenario by Quinn et al. (2016). The advantage of having such a system is that the central image forms further away from the lens centre, where the potential is less steep, so that the central image is less demagnified. The

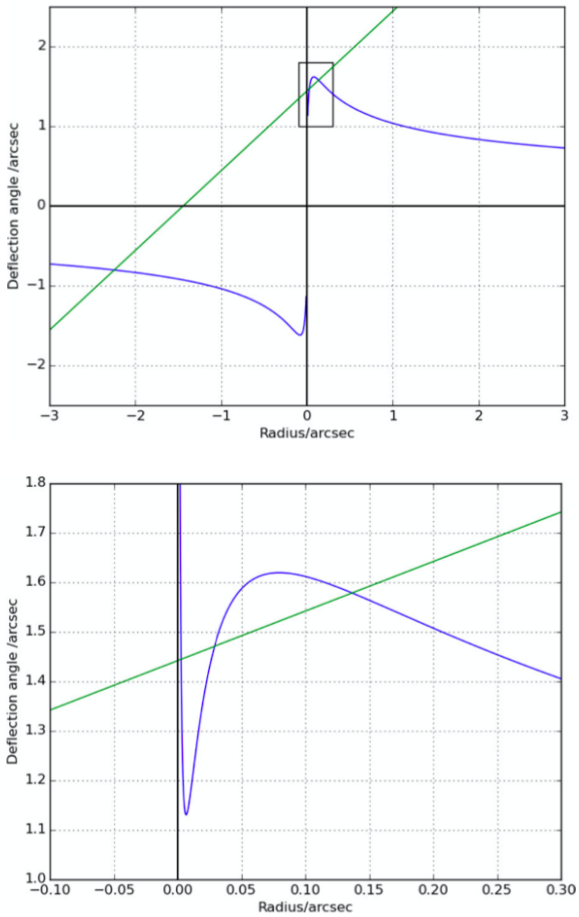


Figure 1.3: An example of a bend angle diagram for a typical galaxy gravitational potential with a central SMBH. The lower panel is the expansion of the upper panel as indicated by the box. The lens equation (green curve) and the bend angle (blue curve) are plotted on the same graph with deflection angle against angular distance from the lens centre. The diagram provides direct graphical solutions, giving image positions at the intersections of the two curves. The effect of the central SMBH is to make a very steep central region in the lens. In this example, the source is located near the Einstein radius, resulting in a central image formed further away from the lens centre, where the potential is less steep. This two-image system provides the best scenario for central image detection. Reproduced from Quinn et al. (2016).

central image is also less obscured by the lensing galaxy, resulting in a higher chance of detection. Since such a configuration gives a larger flux ratio between the two bright components, two-image lens systems with a large flux ratio between the two components provide a higher possibility to detect the central image.

Rusin & Ma (2001), Winn et al. (2003), Wong et al. (2015) have attempted detecting central images to set constraints on the inner mass profiles of galaxies. However, the detections of the central images are extremely difficult. For example, Rusin & Ma (2001) observed six radio-loud lenses in search of the central image but none showed a detection. Nevertheless, the detection limits of these observations allowed them to still set constraints on the inner mass profiles of galaxies. As mentioned, the central image becomes non-detectable once the central mass density of a lensing galaxy exceeds a certain limit. By observing a lens system at a certain noise level, we are able to find the magnification limit of the central component and set constraints on the inner mass profile of the lensing galaxy. Quinn et al. (2016) studied the central image of lens system CLASS B1030+074 quantitatively. Using the Nuker law in 5-dimensional parameter space $(\alpha, \beta, \gamma, r_b, \kappa_b)$ and later adding velocity dispersion σ to it, the source positions and convergence κ were fitted to match with the observed flux ratio between the brightest and central component for a range of r_b and γ values, while the other parameters were set by comparing them to Virgo galaxies with similar Einstein radius.

1.3 Quasars

This section is a review on previous quasar studies relevant to this thesis and describes the motive of such investigations. More detailed discussions on the topic can be found in Blundell & Beasley (1998), Leipski et al. (2006), Condon et al. (2013), Jackson et al. (2015), Hartley et al. (2019).

1.3.1 Radio-loud and Radio-quiet Quasars

Since the discovery of the first quasar (Schmidt 1963), most of the quasars are optically selected and emit a relatively low level of radio signal. By calculating the ratio R between the radio and optical fluxes

$$R = \frac{S_{5\mathrm{GHz}}}{S_{\mathrm{o}}}, \tag{1.25}$$

where $S_{5\mathrm{GHz}}$ is the radio flux density at 5 GHz and S_{o} is the monochromatic flux density at optical B-band, one can separate radio-loud (RLQs) and radio-quiet quasars (RQQs) at $R = 10$. Kellermann et al. (1989) conducted a radio survey observing more than 100 quasars and found that ~90% of quasars are radio-quiet with $R < 1$. However, it is not clear whether RLQs and RQQs are two different classes of objects, or whether the two share the same emission mechanism. Croton et al. (2006) showed that the observed star formation rate of galaxies is lower than expected, so star formation must be quenched. It was a widely accepted theory that the emission mechanism of RLQs is due to

the outflow of materials by an active galactic nucleus (AGN) that produces synchrotron radiation (Cotton et al. 1980). Bi-polar jets have previously been observed to confirm the theory (Figure 1.4; Begelman, Blandford & Rees 1984; Bahcall et al. 1995). On the contrary, RQQs are not associated with such a feature. As a result, different mechanisms have been proposed for the emission mechanism of RQQs. The following sections address the possible emission mechanisms of RQQs and discuss previous literatures on the topic.

1.3.2 AGN Mechanism

It is possible that the same AGN mechanism is at work in both RLQs and RQQs. Blundell & Beasley (1998) and Leipski et al. (2006) have observed jet-like structures in intermediate sources that have $1 < R < 10$ using the Very Long Baseline Array (VLBA) and the Very Large Array (VLA) respectively. Such a jet-like morphology suggests that sources with lower radio flux can also be powered by AGNs and have radio emission due to the outflow of materials, but it has not been observed in sources with $R < 1$. By measuring the flux of a source, one can calculate its brightness temperature (Falcke et al. 1996; Ulvestad et al. 2005)

$$T_B = 1.8 \times 10^9 (1+z) \left(\frac{S_\nu}{1\ \mathrm{mJy}}\right) \left(\frac{\nu}{1\ \mathrm{GHz}}\right)^{-2} \left(\frac{\theta_{\mathrm{maj}}\theta_{\mathrm{min}}}{1\ \mathrm{mas}^2}\right)^{-1}, \tag{1.26}$$

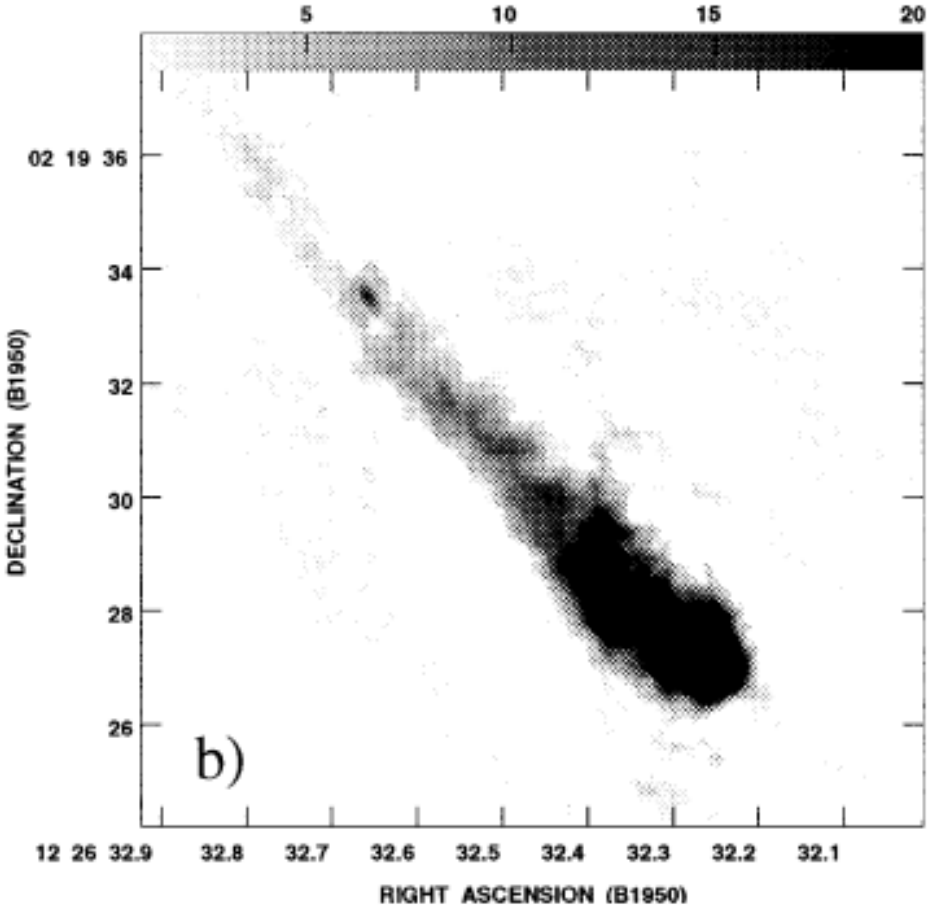


Figure 1.4: Grey-scale image of the jet in quasar 3C 273 with MERLIN at wavelength of 18 cm. The direct imaging of the jet in radio-loud quasars confirmed that RLQs are supported by AGNs. Reproduced from Bahcall et al. (1995).

where S_ν is the flux density of the source at frequency ν, and $\theta_{\rm maj}$ and $\theta_{\rm min}$ are the lengths of the semi-major and semi-minor axes of the observing beam in arcsec. Ulvestad et al. (2005) have observed five RQQs with VLBA to study their emission mechanisms. A jet-like structure is observed in one of the sources, and all the sources have brightness temperatures of at least 10^8 K. As a result, they suggested that RQQs have the same emission mechanisms as RLQs, but at a smaller scale. Hartley et al. (2019) observed the RQQ HS 0810+2554 with the European VLBI Network (EVN) and e-MERLIN, and a source brightness temperature of at least 8.4×10^6 K was obtained. This indicates that the source is powered by an AGN in the galaxy (typically $\sim 10^6 - 10^9$ K). Barvainis et al. (2005) also found that the variability in flux densities of RLQs and RQQs are independent of R, and the radio spectral indices also follow the same pattern. These indicate that the inner regions of all quasars are the same, and have the same emission mechanism. If an RQQ is powered by AGN, only the central region where the SMBH is inhabited will emit radiation and a very compact source can be observed.

The flat spectrum feature of RQQs was explained by Cotton et al. (1980) with the "cosmic conspiracy" core model. The theory states that a series of synchrotron radiations formed by different components in the core superimpose on one another to produce a smooth and flat spectrum. This was a widely accepted theory for decades until more recent studies suggested other possible emission mechanisms, where they are discussed hereafter.

1.3.3 Star Formation

Star formation in the vicinity of the SMBH is another popular hypothesis for the emission mechanism of RQQs. Condon et al. (2013) showed that the 1.4-GHz radio luminosity function of quasars shows an upturn at $L_{1.4\mathrm{GHz}} < 10^{24}$ W Hz^{-1}, as shown in Figure 1.5. The luminosity function of previously studied quasars were compared with the luminosity functions of nearby galaxies powered by AGNs and star formations. Results show that the 1.4-GHz luminosity function of RQQs matches with that of star-forming galaxies. This indicates that the change in number of quasars at different luminosities follows that of star-forming galaxies, and that the emission mechanisms of both are the same. Sopp & Alexander (1991), on the other hand, found that the radio-FIR correlation of RQQs follows that of star-forming galaxies. It is important to note that previous studies supporting star formation as the emission mechanism of RQQs are based on indirect evidence where RQQs and star-forming galaxies show similar properties.

If the emission mechanism of RQQs at work is dominated by star formation around the central SMBH, emission will be more extended and a larger source size will be expected. A low brightness temperature of the source ($< 10^4$ K) will also be expected.

1.3.4 Magnetically Heated Coronae

Laor & Behar (2008) proposed the possibility that the magnetically heated

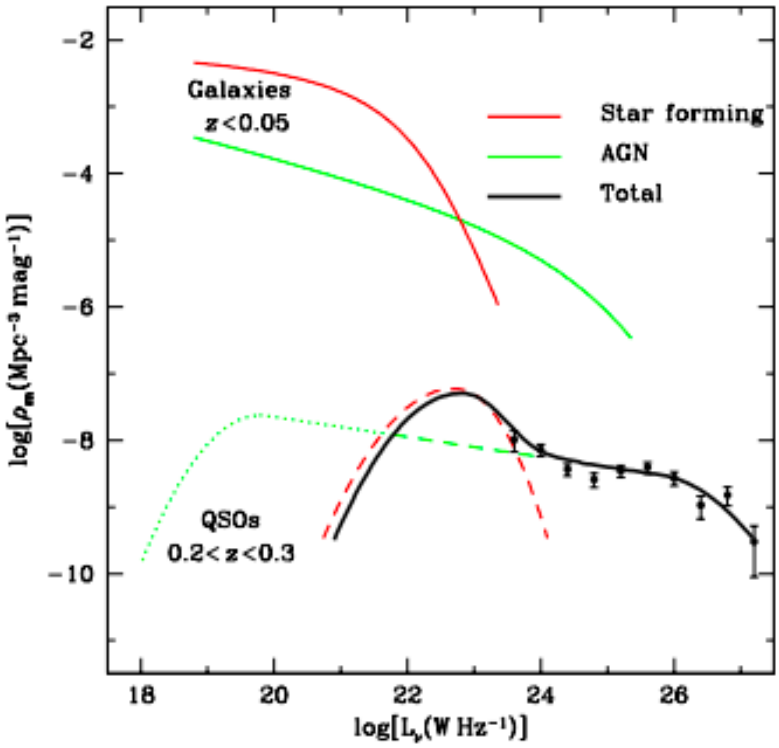


Figure 1.5: The radio luminosity functions of nearby radio galaxies ($z < 0.05$) powered by star formation (continuous red line) and AGN (continuous green line) at 1.4 GHz. The black line represents the luminosity function model $0.2 < z < 0.3$ quasars, red and green dashed line represent the power-law extrapolations for the luminosity function of these quasars if they are only powered by star formation and AGNs respectively. The luminosity function of quasars shows an upturn at $L_{1.4\mathrm{GHz}} < 10^{24}$ W Hz^{-1}, following the red dashed line. This indicates that quasars with lower radio luminosities may be primarily powered by star formation instead of AGNs. Reproduced from Condon et al. (2013).

coronae could explain the radio emission in RQQs. They found a strong correlation between the radio luminosity (L_R) and X-ray luminosity (L_X) in RQQs with $L_R/L_X \sim 10^{-5}$. When compared to the Güdel-Benz relation (Guedel & Benz 1993), which gives a similar ratio for coronally active stars, the correlation matches with the relation. Therefore, they suggested that the radio emission in RQQs comes from a coronal origin, where hot thermal plasma at $T \sim 10^7$ K produces free-free and line emission at X-ray and the non-thermal electrons produce synchrotron radiation at radio wavelengths. Alhosani et al. (2022) found that the radio emission of RQQs is dependent on the Eddington ratio ($r_{\rm Edd} = L_{\rm AGN}/L_{\rm Edd}$), and that the emission has a flat spectrum and comes from a compact region only at low Eddington ratio $r_{\rm Edd} < 3$. As $r_{\rm Edd}$ becomes larger, the spectrum becomes steeper and emission originates from a more diffused area. The result supports the possibility that radio emission in RQQs is due to the inner accretion corona disk, as stellar-mass black holes also show similar properties to depend on the Eddington ratio. Chen et al. (2023) found that the brightness temperatures of 18 RQQs in a sample are lower than 10^7 K, and their L_R/L_X ratio tightly follow the Güdel-Benz relation. As the radio emissions of these sources also has a flat spectrum which indicate compact emission regions, Chen et al. (2023) suggested that magnetically heated coronae are at work in these quasars.

1.3.5 Optically Thin Bremsstrahlung

Blundell & Kuncic (2007) suggested that emission from RQQs is powered

by optically thin bremsstrahlung, due to their flat spectra, along with the unsuccessful attempts to observe their radio jets. They argued that due to the lack of evidence proving the "cosmic conspiracy" theory, the high detected brightness temperatures possibly represent an optically thin thermal emission. When the temperature is high enough ($T \simeq 10^7$ K) to completely ionize gas, free-free emission can result if the plasma is dense enough while being sub-relativistic. Such an optically thin bremsstrahlung emission will give a flat spectrum at radio frequencies that is usually observed in RQQs.

1.3.6 Radiatively Driven Shock Front

Zakamska & Greene (2014) studied the Sloan Digital Sky Survey (SDSS) spectra of RQQs and suggested that the radio emission in RQQs is powered by radiatively driven shock front. By studying the [OIII] emission line, a blueshift feature that represents outflow was observed. The velocity width of the line also shows an outflow velocity of ~ 2000 km s^{-1}. As such features in quasi-neutral medium become more apparent with the increase of the outflow velocity, relativistic particles can be accelerated by the shocks during outflows.

1.3.7 Motivation

While the radio emission mechanism of RQQs remains a puzzle, radio lenses offer a good opportunity to solve the problem. In contrast to optical sources

(~ 1 μas), emission of radio sources comes from a region more extended than the accretion disk, resulting in a larger source size (~ 1 mas). Radio sources are also insensitive to microlensing, as their source characteristic sizes are much bigger than the characteristic sizes of microlensing caustics. This attribute becomes exceptionally helpful when one is interested in investigating the substructures of the lensing galaxy.

Strong lensing, on the other hand, provides the magnification required for RQQ detection. Using radio interferometers to observe the same source with increasing baseline lengths, the resolutions of the observations change, and large-scale structures in the source may become undetected (Section 2.2.2). Non-detection of a source in a long-baseline observation indicates emission from a more extended star-forming region around the central SMBH, while a successful detection indicates a compact source powered by an AGN. This is the most conclusive method to investigate whether radio emission of the quasars are due to star formation or AGN mechanism. Hartley et al. (2019) stated that the radio-FIR correlation may not be able to separate the AGN mechanism from star formation as sources powered by AGNs may not always show the radio excess from the correlation. On the other hand, the spectral index of the source alone is not conclusive to whether AGN mechanism or star formation is at work. This is because steep spectral indices can result from synchrotron radiation produced by the AGN, or supernovae explosion in star-forming regions.

Wucknitz & Volino (2008) have observed the gravitational lens J1131-

1231 with MERLIN and EVN in e-VLBI mode. Observations in MERLIN revealed lensed star-forming regions and the compact cores, but the EVN data with longer baselines showed a non-detection of the AGN in the lensed cores. Therefore, they suggested that J1131-1231 is actually powered by star formation. When the source is not detected in an observation, we can set a lower limit on the source size and the upper limit on the brightness temperature can be calculated using Equation 1.26, where the beam size is obtained from the observation. With a low brightness temperature ($T_B < 10^4$ K), we can conclude that the RQQ is powered by star formation. Conversely, if the source is detected, we can obtain a lower limit on the brightness temperature using the same method. A high brightness temperature ($\sim 10^6 - 10^9$ K) represents that AGN mechanism is responsible for the radio emission of the source.

Chapter 2

Radio Interferometry

2.1 Introduction

The resolution of a single-dish radio telescope can be defined as the full width at half maximum (FWHM) of the main beam. Quantitatively, the angular resolution θ is given by $\theta \sim \lambda/w$, where θ is in units of radians, λ is the observing wavelength and w is the diameter of the telescope. When compared to observations done at shorter wavelengths (e.g. optical), it is apparent that the resolution in the radio regime is far worse. Radio interferometry uses two antennas separated by a baseline distance d and the angular resolution becomes $\theta \sim \lambda/d$ (Thompson et al. 2017). As a result, resolution of an observation can be greatly improved. The idea of radio interferometry is based on Young's slits, where monochromatic coherent light from a single distant point source is diffracted and the interference pattern is observed. Radio interfer-

ometry was first used by Ryle & Vonberg (1946) to detect solar radiation and has then been successful in astronomical studies. For example, the Very Long Baseline Interferometer (VLBI) Event Horizon Telescope (EHT) that observed the central black hole in the M87 galaxy can achieve an angular resolution of 15 μas at 345 GHz (Akiyama et al. 2019).

The observations in this thesis were done with the enhanced Multi Element Remotely Linked Interferometer Network (e-MERLIN) in the United Kingdom. In this introductory chapter of radio interferometry, the techniques of radio interferometry and calibration processes will be discussed.

2.2 Observations

The setup of a radio interferometer with two antennas is shown in Figure 2.1.

Signals in the two antennas experience a geometric time delay of τ, which will be compensated by the electronic delay τ_i produced by the instruments. After that, signals are first multiplied by the local oscillator, then correlated by averaging the signals between two antennas over a short time. A single correlation of signals exists in a form of integral to produce the response function R, which is given by

$$R = e^{ik\vec{B}\cdot\hat{s}} \int I(\hat{\sigma}) e^{ik\vec{b}\cdot\hat{\sigma}}\, d\sigma, \tag{2.1}$$

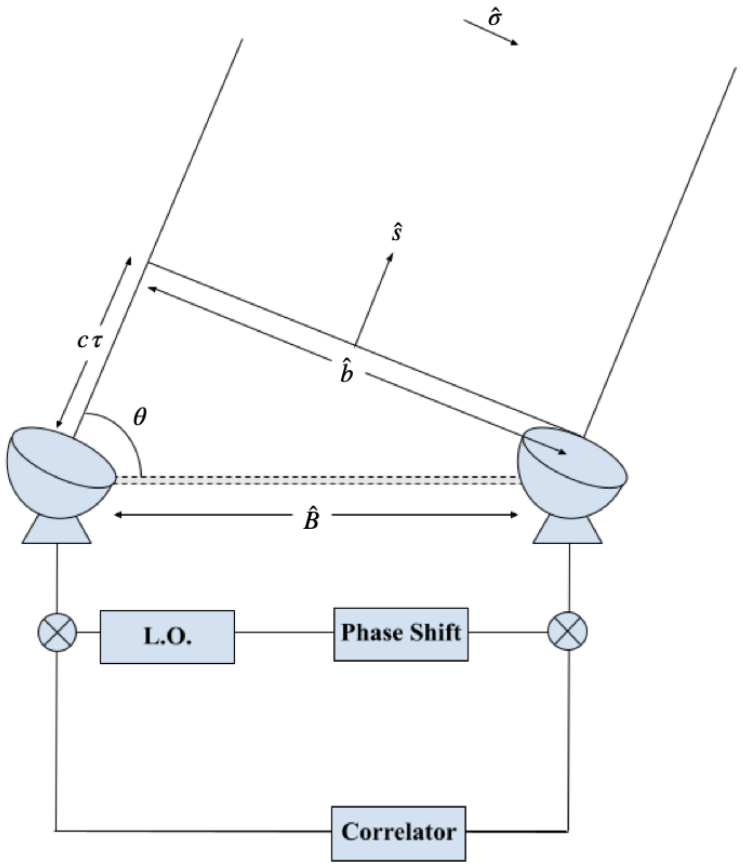


Figure 2.1: Schematic diagram of a two-element radio interferometer with baseline length $\hat{B}$ observing a distant source at angle θ. The unit vectors $\hat{s}$ and $\hat{\sigma}$ represent the observing direction and the moving direction of the source in the sky respectively. Signals in the two antennas experience a geometry delay τ, which will be compensated later by an electronic delay and correlated together.

where $\vec{B}$ is the baseline vector, $\vec{b}$ is the projected baseline vector in the sky, $\hat{s}$ is the unit vector that represents the pointing of the interferometer, and $\hat{\sigma}$ is the unit vector that represents the moving direction of the source. The response function R gives the total signal from the source at a given orientation and baseline length of a pair of telescopes. The exponential term outside the integral, $e^{ik\vec{B}\cdot\hat{s}}$, is the "geometry term", which is only affected by the alignment of the telescopes; the integral, $\int I(\hat{\sigma})e^{ik\vec{b}\cdot\hat{\sigma}}\,d\sigma$ represents the "source term" and is the Fourier Transform equation. This is also known as the Van Cittert-Zernike theorem that describes the relation between the intensity distribution of a source and its complex visibility (Figure 2.2; Burke, Graham-Smith & Wilkinson 2019; Taylor et al. 1999; Jackson 2008).

For any observation on a given source, $\vec{B}$, $\vec{b}$, and $\hat{s}$ vary over time due to rotation of the Earth and movement of the source in the sky. Since the geometric time delay τ is given by $\vec{B}\cos\theta/c$, it also varies with time to produce a fringe rate. Therefore, the electronic time delay τ_i and phase rotation ϕ_f have to be adjusted over time and applied to compensate for the geometric delay. Such a procedure removes the geometry term from the response function, and we are left with the source term only, which gives information about the source. Converting to Cartesian coordinates, we can obtain the response function equation

$$R(u,v) = \iint \frac{I(x,y)}{\sqrt{1-x^2-y^2}} e^{2\pi i(ux+vy)}\,dx\,dy, \tag{2.2}$$

where the response function $R(u, v)$ represents the visibility of the source on the u-v plane and is the Fourier Transform of the sky brightness distribution $I(x, y)$. The coordinate systems (u, v) and (x, y) represent positions on the u-v plane and sky plane respectively, and the u-v plane will be discussed in the next section.

2.2.1 The u-v plane

The u-v plane shows the lengths and orientations of a projected baseline between two telescopes in an observation (with u-axis representing the east-west direction and v-axis representing the north-south direction), where the baseline length is expressed in units of observing wavelength. Due to the rotation of the Earth, a pair of telescopes draws a u-v track on the u-v plane. Given an observation of 24 hours where the source is fully observed, a baseline leaves an ellipse on the u-v plane. The Fourier Transform between the Cartesian coordinates and the u-v plane contains information of the source. For instance, if the target is a double source with separation of 1 rad, the u-v plane will be filled with stripes separated by 1 λ. In observations where the double source is separated by 1 arcsec, the stripes on the u-v plane are separated by $\sim 206000/a$ λ.

2.2.2 Source Size

Source size has a significant effect on the observation due to the properties of interferometry. If the apparent source size exceeds the resolution of the observation, the baselines do not have sufficient sensitivities to detect the large-scale structures in the source, and the source is not recorded on large spatial scales. Such a loss of information in the source is irretrievable, so the observing frequency and baseline lengths in an observation have to be carefully chosen, depending on the source. By observing the same source several times with increasing baseline lengths, we can obtain a lower limit on the apparent source size when the source disappears in a long-baseline observation. There are a few caveats when trying to observe with higher frequencies (higher resolutions) besides the fact that large-scale structures of the source may be lost, they include lower intrinsic flux densities in sources, more serious atmospheric effects, and worse sensitivities in antennas.

2.2.3 Sensitivity

The sensitivity of an observation with an array of telescopes can be given by

$$\Delta S_{rms} = \frac{2k_b T_{sys}}{N_{ant} A_e \sqrt{\Delta\nu\tau}}, \tag{2.3}$$

where ΔS_{rms} is the root-mean-square (rms) noise level of the observation, k_b is the Boltzmann constant, T_{sys} is the system temperature of each antenna,

N_{ant} is the number of antennas, A_e is the effective collecting area of each antenna, $\Delta\nu$ is the total bandwidth of the observation, and τ is the total integration time of the observation. The product of the total bandwidth and the total integration time $\Delta\nu\tau$ is proportional to the number of independent measurements recorded in the observation and the $\tau^{\frac{1}{2}}$ factor assumed trivial systematic errors so that τ will not dominate the error calculation.

2.2.4 Fringe Fitting

Practically, the electronic delay depends on the arrangement of antennas in an array, the position of the source, as well as atmospheric conditions that change constantly. For small arrays of interferometers where the antennas are well-connected and are observing through a relatively small patch of sky, the electronic time delay τ_i and the phase rotation ϕ_f can be adjusted throughout the observation to adapt to the changes and reference the phase centre to the centre of the target source. VLBI observations, on the other hand, face difficulties such as different atmospheric conditions at different telescopes, errors in telescope positions due to tectonic movements, and errors in telescope clocks. On top of that, τ_i and ϕ_f cannot be applied to VLBI data during observations as VLBI data are correlated in a later stage. Fringe fitting is thus introduced to correct for the phases as a function of time and frequency (delay and rate). The antenna phase, delay, and rate errors relative to a reference antenna are estimated using all of the baselines by observing a phase calibrator. The solutions are then applied to the data to correct for

the phases.

2.3 Field of View (FOV)

2.3.1 Bandwidth Smearing

An important assumption during the consideration of Young's slit is that the incoming light is monochromatic, so that only a single wavelength is considered. Such an assumption ensures that the fringe patterns are all in phase and the visibility is at maximum. In practical observations, in order to improve the sensitivity of an observation, continuum observations are often done with a bandwidth $\Delta\nu$ (Equation 2.3), which can result in bandwidth smearing. Light waves at different wavelengths destructively interfere with one another at larger values of delays, causing the visibility to be reduced.

On the u-v plane, u-v tracks of different wavelengths have small differences in radial distance from the phase centre, which can lead to a radial averaging effect when averaging the data. If $\Delta\nu$ reaches approximately the observing frequency ν, bandwidth smearing will bring severe effects to the observations. The FOV affected by bandwidth smearing $\Delta\theta$ is given by (Taylor et al. 1999)

$$\Delta\theta = \frac{\nu}{\Delta\nu} \cdot \frac{\lambda}{d}. \tag{2.4}$$

A common way to alleviate the effect of bandwidth smearing is to divide bandwidths into channels such that each channel width is $\Delta\nu_c$, so that the

FOV becomes (Taylor et al., 1999)

$$\Delta\theta = \frac{\nu}{\Delta\nu_c} \cdot \frac{\lambda}{d}. \tag{2.5}$$

The two observations with the e-MERLIN in this thesis are in different observing frequencies. In the L-band (1.25-1.75 GHz), the total bandwidth of 512 MHz is divided into 8 Intermediate Frequencies (IFs), each of 64 MHz, and are further divided into 128 channels. For the C-band (4.5-7.5 GHz), the total bandwidth of 512 MHz is divided into 4 IFs, each of 128 MHz, and are further divided into 128 channels.

2.3.2 Integration Time Smearing

The integration time helps reduce the size of a dataset by averaging the data within an integration time range. However, since the data on the u-v plane changes with time, a tangential smearing effect that gets worse further away from the phase centre is resulted if the integration time is too large. The FOV affected by integration time smearing $\Delta\theta$ is given by

$$\Delta\theta = \frac{1}{\omega t_{\rm int}} \cdot \frac{\lambda}{d}, \tag{2.6}$$

where ω is the angular frequency of the Earth's rotation and $t_{\rm int}$ is the integration time. Therefore, for an observation of a large FOV, $t_{\rm int}$ has to be small enough to avoid the effect of time smearing.

2.3.3 Wide-field Imaging

When observing a wide-field, the curvature of the sky plane can impose a geometrical smearing effect on the image due to the vertical displacement. Therefore, to correct for the phase difference in this direction, the "w-term" is introduced. The response function becomes

$$R(u,v,w) = \iint \frac{I(x,y)}{\sqrt{1-x^2-y^2}} e^{2\pi i(ux+vy)} e^{2\pi iw(\sqrt{1-x^2-y^2}-1)}\, dx\, dy, \tag{2.7}$$

where the extra phase term is represented by $e^{2\pi iw(\sqrt{1-x^2-y^2}-1)}$ and w is the "w-term". As w is position dependent, any baseline with N-S extension will have $w \neq 0$. A standard solution to the w-term is to use w-projection to separately Fourier Transform the extra phase term to get the w-kernel $\tilde{G}(u,v,w)$. Using the inverse of the w-kernel $\tilde{G}(u,v,-w)$, we can carry out the convolution

$$V(u,v,w=0) = V(u,v,w) * \tilde{G}(u,v,-w)$$

to find the projection of the image. Surveys that explore a large part of the sky are usually affected by the 'w-term'. The observations in this thesis only observes individual sources and do not have a large FOV, so they are not impacted by the curvature of the sky plane. Therefore, the 'w-term' is not considered.

2.4 Imaging

2.4.1 Deconvolution

As mentioned in Section 2.2, if the observation covers the whole u-v plane, the response function on the u-v plane $R(u, v)$ is simply given by the Fourier Transformation of the true sky brightness distribution $I(x, y)$ (Equation 2.2). However, since the real-time u-v coverage is never perfect, and is given by the sampling function $S(u, v)$ which has a value of 1 at the sampled coordinates and 0 at non-sampled coordinates, the real-time response function on the u-v plane is given by

$$R_D(u, v) = R(u, v) \times S(u, v), \tag{2.8}$$

where $R_D(u, v)$ is the real-time visibility on the u-v plane. By the Convolution Theorem, Equation 2.8 can be written as

$$I_D(x, y) = I(x, y) * B(x, y), \tag{2.9}$$

where $I_D(x, y)$ is the dirty image of the observation and $B(x, y)$ is the dirty beam, which is the Fourier Transform of the sampling function. As a result, the clean image $I(x, y)$ can be obtained using deconvolution of the dirty image.

2.4.2 CLEAN

The method used in deconvolving dirty images in this dissertation is the `CLEAN` algorithm (Högbom 1974). The algorithm chooses the brightest pixel in the dirty map as the phase centre and subtracts the dirty beam from the image after scaling. A residual map which consists of only noise is constructed after subtraction of the dirty beam. The subtracted map will then be convolved with a restoring beam and added back to the residual map to produce an image after each iteration. During each `CLEAN` algorithm, the pixel size can be chosen for the reconstruction of the map. It has to be carefully chosen to approximately one-third of the resolution of the observation such that it will not be too small that causes the computation time to be too long, nor it will not be too large that the resolution of the map and the sampling of data are affected. The other factor that affects the quality of a map is the weighting of the `CLEAN` process. Data points are usually concentrated towards the centre of the u-v plane due to a more central u-v coverage of shorter baselines, and a larger number of shorter baselines in a practical observation. "Natural weighting" does not apply any weight to the visibility data, and therefore produces an image with higher signal-to-noise ratio (SNR), but a worse resolution since the Fourier Transform of a more centrally concentrated visibility gives a larger dirty beam. "Uniform weighting", on the other hand, averages out grids in the u-v plane such that each grid is equally weighted. Such a weighting method improves the resolution,

but the SNR becomes worse because of the down-weight of data at the central region of the u-v plane. Briggs (1995) proposed the "Briggs weighting" that consists of a robust value to compromise between the "natural" and "uniform" weighting methods.

2.5 Calibration

The phase and amplitude corruptions caused by the atmosphere change with time, the observed patch of sky, and frequency. Charged particles in the ionosphere and water vapour are responsible for the shifting of phases at low and high frequencies respectively. To correct for the the observed sky visibility ($V_{ij}^{\rm obs}$) between antennas i and j, we have

$$V_{ij}^{\rm obs} = g_i g_j^* V_{ij}^{\rm real}, \tag{2.10}$$

where g_i and g_j are the complex gains for the two antennas respectively, and $V_{ij}^{\rm real}$ is the true sky visibility. With both amplitude and phase solutions included in the complex gains, we can solve Equation 2.10 and obtain the true sky visibility.

2.5.1 Amplitude Calibration

The overall flux density scale is found by observing a source of known flux. Amplitude calibration can then be done by applying the overall flux density

scale to the observed visibilities of the target source. The flux calibrator has to be unresolved in all baselines and its intrinsic flux has to be constant throughout the observation. There are a number of well-studied sources that are available to use as a flux calibrator (Baars et al. 1977; Ott et al. 1994), including 3C286, which was observed in both of the observations in this thesis.

2.5.2 Phase Calibration

Phase calibration, unlike amplitude calibration, is much more challenging because the phase of a source varies on a much shorter timescale than the amplitude. A source in the same isoplanatic patch of the sky as the target source and of known structure has to be used as the phase calibrator. The phase calibrator is observed periodically throughout the observation within a coherence time of the atmosphere, which depends on the observing frequency. By Fourier Transforming the source structure and removing it from the visibility of the phase calibrator, the residual phase structure is interpolated and removed from the response function of the target source. This technique is also called 'phase referencing', and is a very successful method in VLBI (Walker 1999), especially when the target source is too weak or too complex.

2.5.3 Self-calibration

For bright and simple sources, closure quantities are particularly useful when dealing with atmospheric errors. The phase and amplitude errors between antennas are used to set constraints on the phases and amplitudes of the target source. Yet, closure quantities are only able to eliminate atmospheric errors in observations, but not the residual systemic errors in telescopes.

Self-calibration, therefore, can improve the quality of an image by using a model of the sky to solve for the residual phase corruptions with least squares fitting and applying the solutions to the measured data (Cornwell & Wilkinson 1981). This process can be carried out multiple times as the model of the sky can be used again in the next cycle of self-calibration to solve for the phase errors. Phase and amplitude self-calibration can be done separately, with the choice of selecting a solution interval. The usual practice is to correct for the phases before continuing on the amplitude, since the phase corruption is often more severe than the amplitude corruption. It is also important to use a solution interval shorter than the phase coherence time to ensure that the phase does not vary significantly over one solution interval.

Since the number of free parameters to solve for are proportional to the number of telescopes N, and the number of constraints in self-calibration is proportional to the number of baselines $\frac{N(N-1)}{2}$, the number of constraints is larger than the number of free parameters. This makes self-calibration a

sensible procedure. However, it is important to ensure that each antenna has SNR > 3 on the target source in one coherence time to ensure that least squares fitting works well.

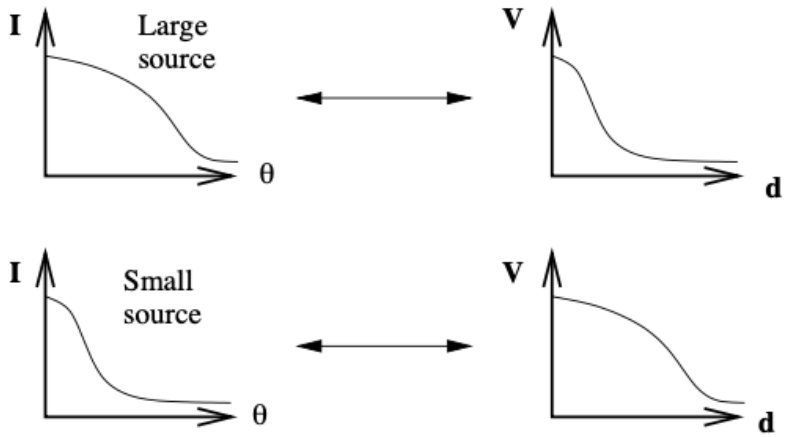


Figure 2.2: Van Cittert-Zernike Theorem showing the Fourier Transform of a large Gaussian results in a small Gaussian, and vice versa. The left panels show the intensity of a source (I) as a function of angular distance (θ); while the right panels show the visibility of interference patterns (V) as a function of slit separation (d). Reproduced from Jackson (2008).

Chapter 3

Radio Observation of CLASS B0850+054

3.1 Introduction

The gravitational lens system CLASS B0850+054 was discovered in the Cosmic Lens All-Sky Survey (CLASS) (Biggs et al. 2003; Browne et al. 2003; Myers et al. 2003). CLASS is a survey to identify radio-loud lens systems by observing more than 10,000 flat-spectrum sources with flux densities greater than 30 mJy at 5 GHz. B0850+054 was confirmed to be a two-image lens system with the VLA 30-s snapshot at 8.5 GHz. It consists of two compact components with a separation of 0.7 arcsec and a flux density ratio of $\sim 6:1$. McKean et al. (2004) measured the spectroscopic redshift for the lensing galaxy $z_l = 0.59$ using the W. M. Keck I telescope, detecting six absorption

features and the 4000 – Å break. A strong emission line at 5995 Å that is related to the background source was also detected. However, it is not clear whether this line corresponds to MgII or Lyα emission, giving source redshifts of z_s = 1.14 and 3.93 respectively. Follow-up observations were done using the VLA, the MERLIN, the VLBA and the Westerbork Synthesis Radio Telescope (WSRT). The VLA 5-GHz, VLA 8.5-GHz, VLA 15-GHz, and the MERLIN 5-GHz maps of B0850+054 (Biggs et al. 2003) are shown in Figure 3.1.

Being a two-image lens system with high flux ratio between the lensed components, B0850+054 is a suitable candidate for central image investigations (Quinn et al. 2016). A previous study (Boyce et al. 2005) tried to detect the central image of B0850+054 and model the lensing galaxy (no data was published unfortunately). To try detecting the central image and determine better constraints for the mass profile of the lensing galaxy, e-MERLIN observations of B0850+054 were proposed at C-Band (5 GHz). The e-MERLIN consists of seven antennas in the United Kingdom (including the 76-m Lovell Telescope at Jodrell Bank Observatory) with a maximum baseline length of ~ 220 km. The observations on B0850+054 is part of the e-MERLIN Legacy Programme (Gravitational Lensing and Galaxy Evolution with e-MERLIN; Jackson, Serjeant & Keynes n.d.) to work on deep mapping of known radio lenses. The sufficiently high sensitivity of e-MERLIN provides a good chance of detecting central images, thus benefits the investigations on mass distributions in the central regions of galaxies.

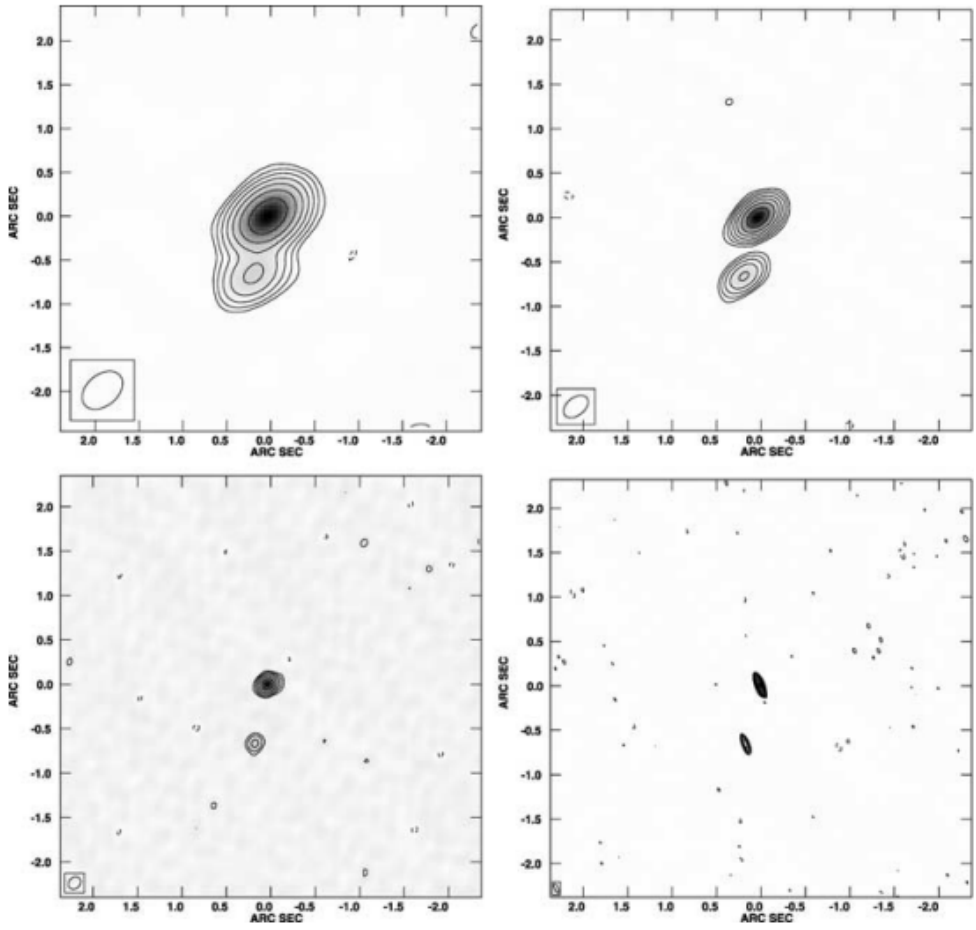


Figure 3.1: Radio maps of B0850+054. VLA 5-GHz map (top left); VLA 8.5-GHz map (top right); VLA 15-GHz map (bottom left); MERLIN 5-GHz map (bottom right). The rms noise level in the MERLIN 5-GHz map is 100 μJy beam^{-1}. With the new e-MERLIN data, the rms noise level can be improved by an order-of-magnitude. Reproduced from Biggs et al. (2003).

In this chapter, the new e-MERLIN data of B0850+054 are presented. The calibration processes are discussed and the results are compared to previous observations. We also present the lens modelling of the data and show new results of the constraints on the lensing galaxy. The results are compared to previous central image investigations such as Norbury (2002), Keeton (2003), Quinn et al. (2016).

3.2 Observations

Four observations on B0850+054 were done in November and December, 2022 over a course of 15 days, using e-MERLIN at C-band (5 GHz). The C-band observation has a frequency range of 4.82 – 5.33 GHz, and a bandwidth of 512 MHz. The bandwidth can be divided into 4 spectral windows (IFs), each with a width of 128 MHz and are further divided into 128 channels. With 4 polarizations and a channel width of 250 kHz, the bandwidth smearing effect can be minimised. Moreover, the average integration time is set to 1 s to minimise the time smearing effect.

The integration time of each observations was ~ 10 hours. The data was initially calibrated with the e-MERLIN CASA pipeline (Moldon 2021) and distributed as part of the e-MERLIN Legacy Programme. The pipeline failed when calibrating data in one of the four observations and was redone. Observations were done on the target source (B0850+054) and a nearby phase-reference source (J0853+0654) alternately on a cycle time of ~ 25 min-

utes and ~ 4 minutes respectively. A bandpass calibrator (J1407+2827) was observed to correct for the errors within a bandpass and a flux calibrator (3C286) was also observed to determine the flux density scale. Since the target source is bright enough and has a simple structure, self-calibration has been done to determine the solutions for the final map of the source. The total on-source integration time was ~ 28 hours, providing a good u-v coverage on the source (Figure 3.2).

3.3 Data Reduction

The calibrated data were examined and calibrated using the Astronomical Image Processing System AIPS[1] (Greisen 1990). The datasets were loaded into AIPS using the task FITLD. Using the task LISTR, the scan lists of the observations were examined. After that, the task SPFLG was used to inspect the u-v data carefully. Colour plots of amplitudes or phases in each baseline with frequency channel on the x-axis and time on the y-axis were plotted. As the pipeline calibration did not remove all the bad data, further calibration was required. Remaining data of bad quality were flagged (deleted) using the interactive TV display in SPFLG. After checking the data of all the sources in every baseline, power spectra of the sources were plotted with the task POSSM. By inspecting the spectrum of each baseline, remaining bad data were then further flagged using the task UVFLG. The e-MERLIN C-Band

[1]Developed by the National Radio Astronomy Observatory (NRAO)

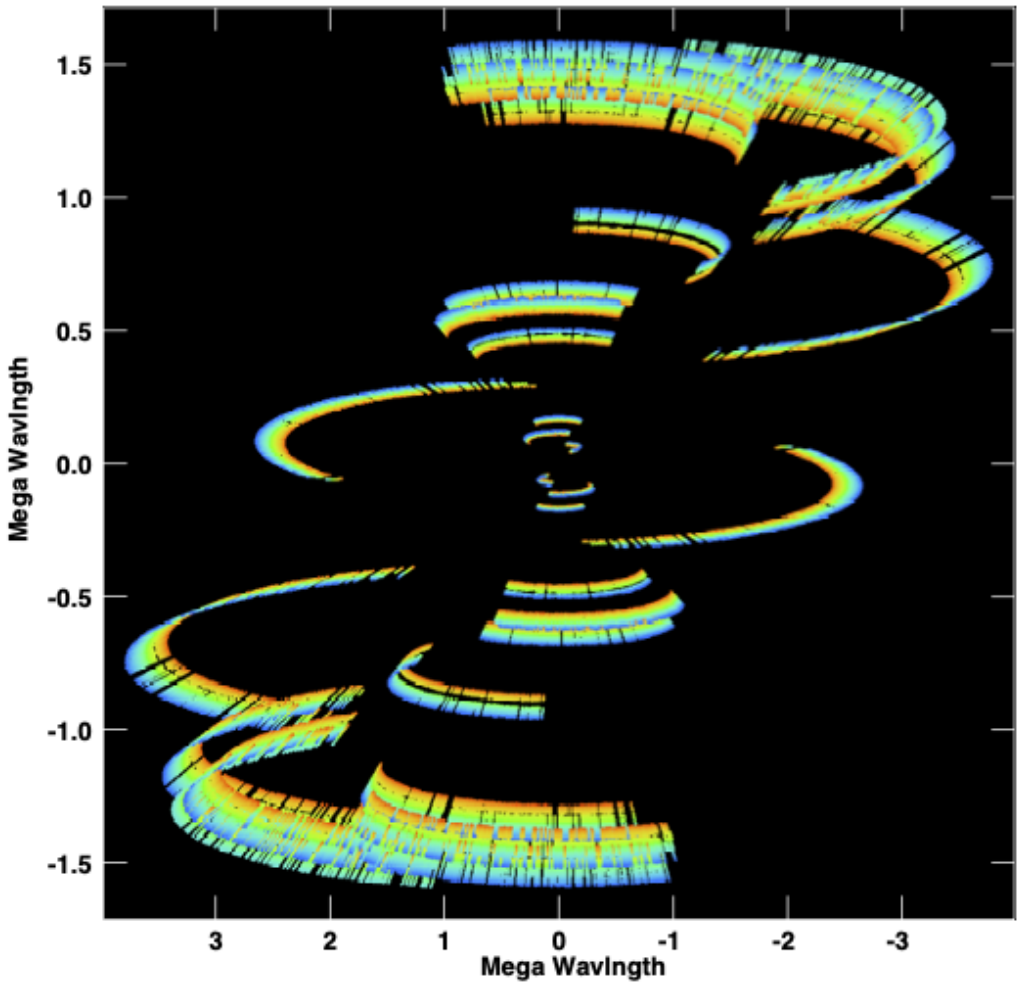


Figure 3.2: The u-v coverage of the observations on B0850+054. Units of the u-v plane are in mega wavelengths, which correspond to 60 km in e-MERLIN 5-GHz observations. Colours on the plot represent the observed frequencies, with red indicating the lowest frequency and blue indicating the highest frequency.

has a certain amount of radio frequency interference (RFI) that generate unwanted signals in the observations. In the observations, the amount of flagged data was relatively small ($< 5\%$) compared to the total amount of data and therefore did not affect the sensitivity of the observations. Due to significant RFI, 15 channels in the IF2 (second spectral window) of all baselines were flagged out. The remaining flags were composed of strong momentary noise or weak RFI that were not flagged during pipeline calibrations.

As mentioned in Section 3.2, the flux scale is usually determined by carrying out gain calibration on standard flux calibrators (e.g. 3C286). However, the model for 3C286 could not be found in AIPS so that the flux density scale could not be determined and transferred to the sources. Therefore, an alternative approach was taken. The flux of the bandpass calibrator (J1407+2827) was measured with POSSM and its spectral index was calculated. After comparing them to the calibrator's absolute flux scale in previous surveys at 5 GHz (Langston et al. 1990; Helmboldt et al. 2007), the flux density scale was set using the task SETJY. The spectral index of a source is given by

$$\log\left(\frac{S_1}{S_2}\right) = -\alpha \log\left(\frac{f_1}{f_2}\right), \tag{3.1}$$

where S_1 and S_2 are the fluxes of the source at frequencies f_1 and f_2 respectively. Since B0852+0515 has a flat spectrum at 5 GHz and J1407+2827 is stable in brightness (Dallacasa et al. 2000), the flux density scale can be determined by this approach. The task BPASS was then used to create band-

pass response functions of the antennas for the error corrections within a bandpass.

3.4 Imaging

An iterative cycle of CLEAN and self-calibration was used to produce the final map of B0850+054. To start the CLEAN process, the task IMAGR was used to map the calibrated dataset. The cell size of the map was set to be 0.015 arcsec × 0.015 arcsec to compromise between computation time and the proper sampling of data. The size of the image was set to be 512 pixels × 512 pixels, and the number of iterations in each CLEAN cycle was set to be 5000. After the first CLEANing, the task CALIB was used to determine solutions of the target source by self-calibration. Figure 3.3 shows an example of the phase solutions obtained from the task CALIB with the Lovell Telescope as the reference antenna. The Lovell Telescope provides the highest sensitivity among all the antennas in e-MERLIN and therefore is reliable as a reference antenna. The task PRTCC was used to obtain the number of CLEAN components in the map up to the first negative value, which was used as a model for the next cycle to avoid self-calibrating against an incorrect model.

Phase self-calibration with a long solution interval (1 hour) was initially carried out before gradually reducing the solution intervals to 10 minutes. A solution interval of 1 hour did not affect the image a lot as the interval was longer than the phase coherence time. It is important to use a short enough

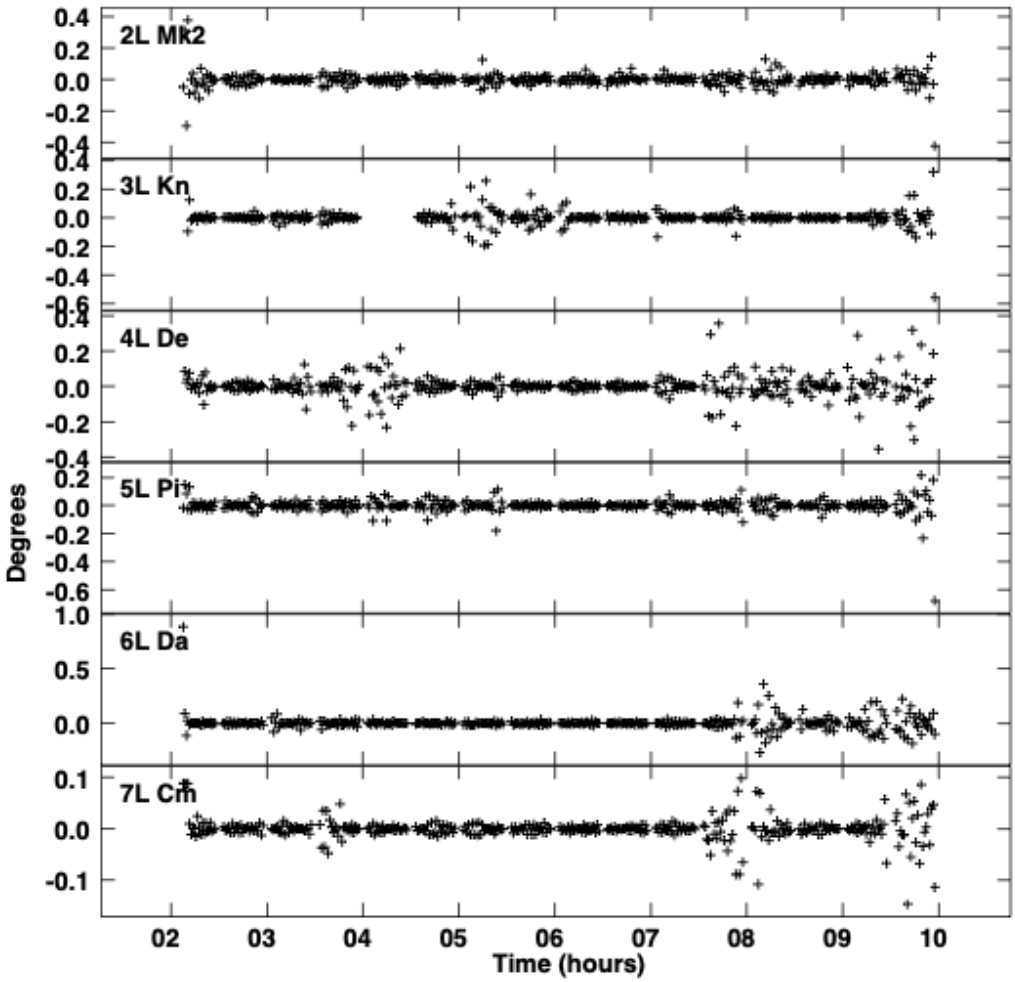


Figure 3.3: Phase solutions obtained from one of the observations with degrees plotted against times. The panels show the phase corrections made to the data points in each antenna. The Lovell Telescope at Jodrell Bank Observatory is taken as the reference antenna.

solution interval for phase self-calibration to obtain accurate solutions as phase varies on a much shorter timescale than amplitude. The task CLCAL was then used to apply the solutions to the the data in a calibration table. The task INDXR was also used to create a blank calibration table to be written with CLCAL. The interval of the table was set to be 0.1 minute to ensure an interval short enough so that all solutions to the phase variation were interpolated to the calibration table. The output calibration table was used to correct the data during the next `CLEAN` cycle.

After phase self-calibration, a cycle of amplitude self-calibration with a solution interval of 15 minutes was carried out to correct for any residual errors. The solution interval of amplitude self-calibration does not have to be as short as the final solution interval of phase self-calibration as the amplitudes are more stable, so that a longer solution interval is already sufficiently short enough for obtaining the true solution. In addition, a solution interval that is too short can result in false solutions as there are too many free parameters. After amplitude self-calibration, a final map was obtained for an observation. Table 3.1 shows the beam sizes and the position angles of each of the observation. The data of all the observations were combined using the tasks SPLIT and DBCON. SPLIT first converted a multi-source u-v dataset into a single source dataset so that the data size was not too large. While ignoring the fully flagged data, it also applied all the calibrations to the target source. DBCON was then used to concatenate the u-v datasets. Since only two u-v datasets could be concatenated at one time, DBCON had to be used

repeatedly to concatenate all u-v datsets. The final map of the target source was constructed using a restoring beam size of 124×32.8 mas^2 at a position angle of 23.76° (Figure 3.4). The restoring beam size and position angle were determined by taking the average of the `CLEAN` beams of the four observations. A robust weighting (Briggs 1995) of zero robustness was used when producing the image to compromise between natural and uniform weighting (Section 2.4.2). The rms noise level of the map is 6.76 μJy beam^{-1}, more than a factor of 10 better than that of the MERLIN 5-GHz map (Biggs et al. 2003). The estimated sensitivity of an e-MERLIN 5-GHz observation with $\sim$ 28 hours on-source time is $\sim$ 5.3 μJy beam^{-1}. The higher rms noise level in the actual observation is due to a noticeable amount of RFI in the C-band, and a number of data that have been flagged out. The artifacts along the declination axis in Figure 3.4 is due to the low elevation of the target source. At low elevation, the u-v coverage becomes highly elliptical along the E-W direction (Figure 3.2). The Fourier Transform of such a u-v coverage results in vertical artifacts around every bright spot in the observed field, as seen in the image. In addition, observing a low declination target source also causes the beam to be highly elliptical.

Flux densities and peak intensities of the two components are listed in Table 3.2. The task JMFIT was used to obtain information about the components. It used least squares fitting to fit 2-dimensional Gaussians to the components in the image.

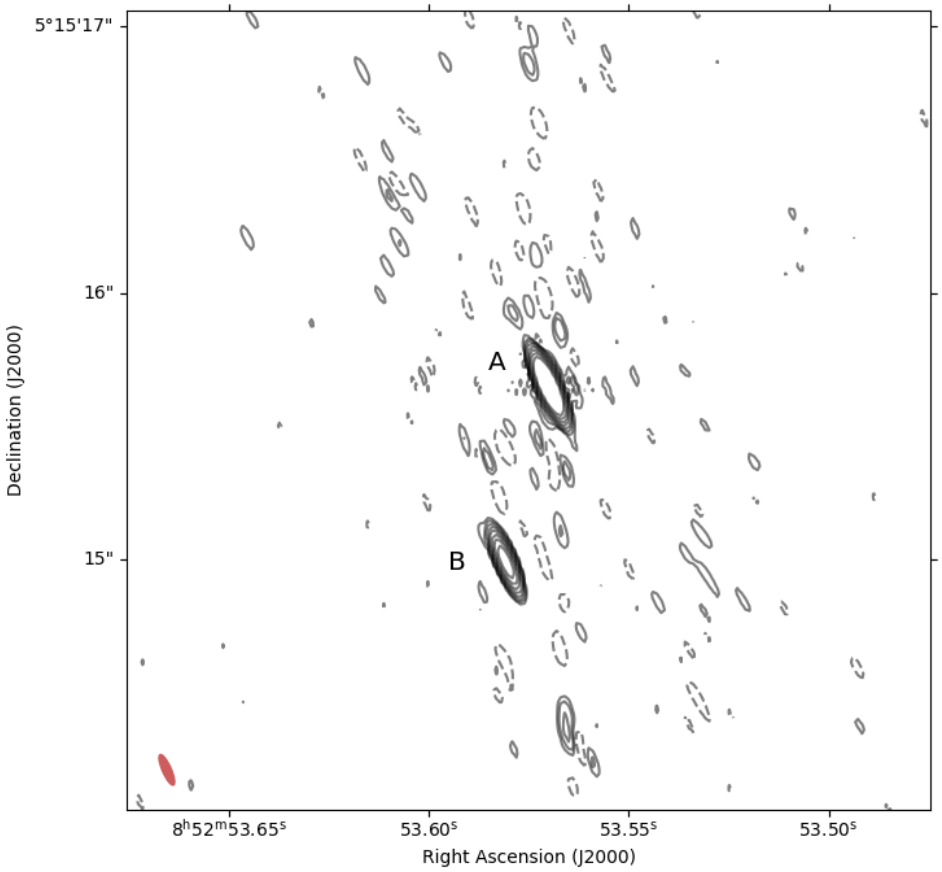


Figure 3.4: e-MERLIN 5-GHz map of B0850+054 with a restoring beam of 124×32.8 mas^2 at a position angle of 23.76°. The data are weighted with a robust weighting with zero robustness (Briggs 1995) and the rms noise level is 6.76 μJy beam^{-1}. The restoring beam is plotted in red at the bottom left corner. Contours are plotted at multiples (-1,1,2,4,8,16,32,64,128) of 3σ where σ is the off-source rms noise level in the map. Components A and B are labelled in the plot.

Obs. Date	bmaj (arcsec)	bmin (arcsec)	bpa (deg)
Nov. 30	0.125	0.033	23.33
Dec. 01	0.128	0.033	24.17
Dec. 12	0.122	0.032	23.41
Dec. 14	0.121	0.034	23.76
Average	0.124	0.033	23.76

Table 3.1: Beam sizes and position angles of the B0850+054 observations. The observation dates, beam major axes, beam minor axes and beam position angles are listed from left to right. The averages of these values are the size and position angle of the restoring beam used to produce the final map.

Cmpt.	Flux density (mJy)	Peak intensity (mJy)
A	45.49 ± 0.02	44.65 ± 0.01
B	7.39 ± 0.02	7.28 ± 0.01

Table 3.2: Flux densities and peak intensities of components A and B as labelled in Figure 3.4.

3.5 Lens Modelling

3.5.1 Data Constraints

A number of constraints are taken from the e-MERLIN 5-GHz map to model the lensing galaxy. The two components give a flux ratio of 6.15:1 ± 0.01:1 and a separation of 0.678±0.002 arcsec that match with previous studies (e.g. Biggs et al. 2003). Owing to the non-detection of the central image in the map, the 3-σ rms noise level also sets a limit on its maximum flux density at 20.28 μJy beam^{-1}, or a limit of ~2250:1 on the flux ratio between the brightest component (Component A) and the central image (Component C).

Expressing the magnification between components A and C in logarithms, we have $\log(\mu_C/\mu_A) \sim -3.35$. Usually, the separations between components and the centre of the lens are also considered, but they are not considered in the modelling of B0850+054 due to the lack of information.

3.5.2 Modelling Strategy

The modelling strategy of this thesis follows the procedures in Keeton (2003) and Quinn et al. (2016) closely, where mass distributions in the form of Nuker law (Equation 1.24; Byun et al. 1996; Faber et al. 1996) are used. Since the inner regions of nearby galaxies are dominated by baryons, the light distributions and mass distributions are similar and can be used interchangeably. The deflection angle of a circularly symmetric lens with a Nuker-law profile is given by Keeton (2003)

$$\alpha_{\rm gal}(R) = \frac{2^{1+\frac{(\beta-\gamma)}{\alpha}}}{2-\gamma}\kappa_b r_b \left(\frac{R}{r_b}\right)^{(1-\gamma)} \times {}_2F_1\left[\frac{2-\gamma}{\alpha}, \frac{\beta-\gamma}{\alpha}, 1+\frac{2-\gamma}{\alpha}, -\left(\frac{R}{r_b}\right)^{\alpha}\right] + \frac{R_E^2}{R}, \quad (3.2)$$

where ${}_2F_1$ is a hypergeometric function, γ and β are the inner and outer power-law indices respectively, r_b is the break radius, α is the sharpness of the break, and I_b is the surface brightness at the break radius. The convergence term κ_b gives us information about the mass-to-light ratio Υ, such that $\kappa_b = \Upsilon I_b/\Sigma_{cr}$ and the surface mass density of the lens at the break radius is ΥI_b.

The last term R_E^2/R is added to the deflection angle when the effect of a black hole is considered, where R_E is the Einstein radius that is proportional to $\mathrm{M}_{BH}^{1/2}$. If a black hole exists in the lensing galaxy, it can heavily demagnify the central image to a flux density below the detection limit of the observation. Norbury (2002) showed that the core radius and the black hole mass affect the detection of a central image. The central image can be completely suppressed at small core radii, or by the presence of a black hole.

The Nuker law has five variables $(\alpha, \beta, \gamma, r_b, \kappa_b)$ that control the profile of the lensing galaxy, and an extra parameter describing the central SMBH can be added. The central SMBH can be expressed in terms of the black hole mass M_{BH} or its velocity dispersion σ, as given in Equation 1.22 (Gebhardt et al. 2000). With the observed image flux ratio and separation as constraints, the Nuker law is optimized by minimising the χ^2 in the image plane. In this 6-parameter space $(\alpha, \beta, \gamma, r_b, \kappa_b, \sigma)$, a grid search is performed by varying the inner power-law index γ and break radius r_b at a range of values. By optimizing the values of the convergence κ_b and the source position s from the lens centre, we try to reproduce the observed flux ratio and separation between the two components, A and B. Using the detection limit of an observation as constraint for the brightness of the non-detected central image to carry out mass modelling allows a straightforward investigation and was done in previous works (e.g. Rusin & Ma 2001; Winn et al. 2003; Quinn et al. 2016).

Ellipticity is not considered in the models due to several reasons: (1)

consistency of models in previous studies (e.g. Keeton 2003; Quinn et al. 2016); (2) elliptical lenses require more constraints than a two-image lens system can provide; (3) the central image magnification is hardly affected by the ellipticity of the lens (Keeton 2003). Keeton (2003) studied how the ellipticity of a lens galaxy would affect the central image magnification and found that by changing the ellipticity of lenses from e = 0 to e = 0.5, the central image magnification distribution did not change. At the same time, while ellipticity played a role in the separation between four-image and two-image lens systems, the central image magnifications of two-image lens systems were less affected by ellipticity and modelling could be done with a circular lens galaxy.

3.6 Early-type Galaxies

Keeton (2003) predicted the core image magnifications of distant galaxies ($z \sim 0.3-1$) by using Nuker-law models derived from nearby early-type galaxies. A sample of 73 elliptical galaxies observed with high-resolution Hubble Space Telescope (HST) imaging at optical or near-IR wavelengths was assembled (e.g. Faber et al. 1996; Carollo et al. 1998; Ravindranath et al. 2001). These galaxies also include measured velocity dispersions σ that represent their central SMBHs. This sample consists of mainly two distinct classes of galaxies – galaxies with shallow inner cores of $\gamma \leq 0.3$ and steep outer profiles (core galaxies); and galaxies that roughly follow power-law, having steeper

inner cores of $\gamma \geq 0.5$ (power-law galaxies). Core galaxies are usually slow rotating elliptical systems that are bright, while power-law galaxies are disky isophotes that are dimmer and with a shorter rotation period (Keeton 2003).

Using the parameters provided in each of the galaxy in the sample, we attempted to reproduce the observed flux ratio and separation between the bright components. Assuming that the galaxies in the Keeton (2003) sample are a good representation of the properties of elliptical galaxies at higher redshifts, we conducted mass modelling to identify the galaxies that are able to reproduce the observational data in B0850+054. The distance of the source from the lens centre was allowed to vary from 0 to 5 arcsec at a step of 0.005 arcsec, and modelling was done with and without a black hole, where the black hole mass was taken from Keeton (2003). The critical surface density is given by $\Sigma_{cr} = (c^2 D_{os})/(4\pi G D_{ol} D_{ls})$, where D_{os}, D_{ol} and D_{ls} are the angular diameter distances between the observer and the source, the observer and the lens, and the lens and the sources respectively. Therefore, redshifts of the lens and the source both have an effect on the lensing equation and the convergence of the lens. We corrected the lens redshift z_l and source redshift z_s from $(z_l, z_s) = (0.59, 1.14)$ to $(0.5, 0.2)$ to follow the assumptions made in Keeton (2003) models, where we took $z_s = 1.14$ as the true redshift of the background source.

Only one of the galaxies (M87) in the Keeton (2003) sample is able to reconstruct the observed flux ratio and separation with a χ^2 value smaller than 5. Figure 3.5 shows the bend angle diagrams of this model, with and

without a central SMBH respectively. The Nuker-law parameters of the modelled galaxy are listed in the titles of the bend angle diagrams in Figure 3.5. The galaxy is a core galaxy with a small value of γ = 0.14 and a steeper outer profile of β = 1.33. M87 has an Einstein radius of 0.32 arcsec and a radial critical radius of 0.008 arcsec. The image separation is successfully reproduced at 0.67 arcsec regardless of whether the black hole is included, and the reproduced image flux ratio without a black hole in logarithm is $\log(\mu_C/\mu_A) = -5.03$. Therefore, if the central SMBH is not considered, we would expect to detect a central image above the detection limit of e-MERLIN at 5 GHz. The assumption in such a consideration is that this galaxy is identical to the lensing galaxy responsible for the lensing of the background quasar in B0850+054. However, when the central SMBH is included, the logarithm of the reproduced image flux ratio is $\log(\mu_C/\mu_A) = -21.14$, which is far below the detection limit of our observation. As a result, it is expected that no central image will be detected if the galaxy is identical to M87.

There are several reasons why there is only one galaxy that is able to reproduce the observed flux ratio and separation. Firstly, the sample size of the Keeton (2003) sample is not large enough to illustrate different types of galaxies that can reproduce the observed data. Secondly, the sample is vulnerable to selection bias. Galaxies in the sample were observed with HST imaging at optical and near-IR wavelengths and are nearby early-type galaxies. Therefore, these galaxies are usually less massive and have a smaller

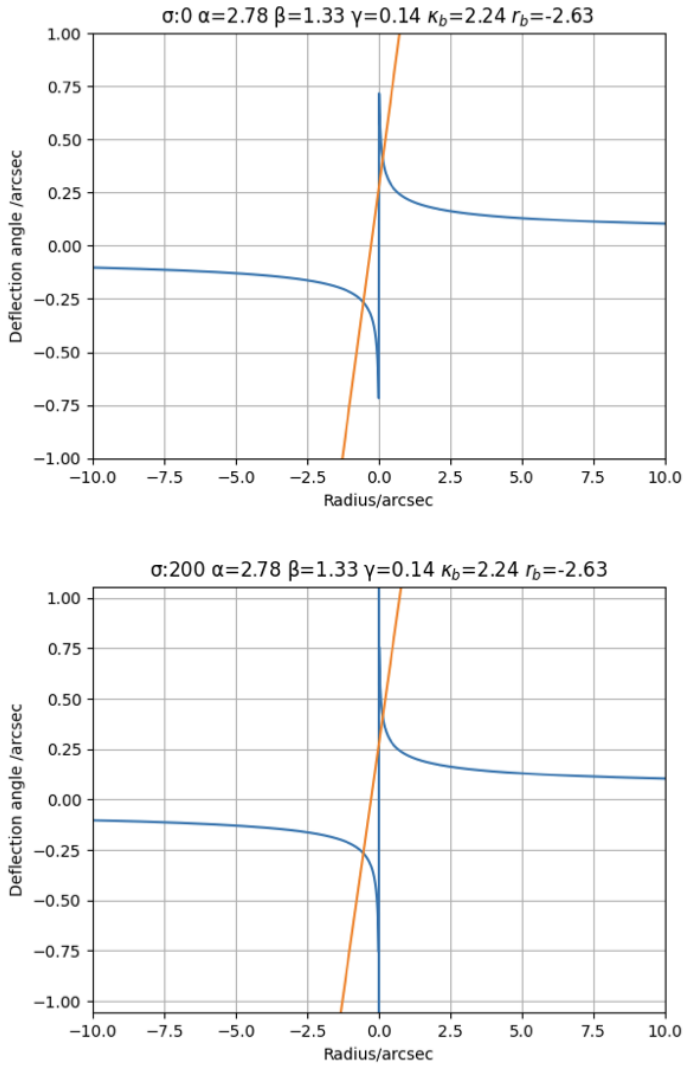


Figure 3.5: Bend angle diagrams with M87 as the lensing galaxy that are able to reproduce the observed source separation and flux ratios in B0850+054. Models with (bottom) and without (top) a black hole are considered. The Nuker-law parameters of M87 are listed on top of each diagram.

Einstein radius. Galaxies with sub-arcsec scale Einstein radius are much less likely to produce images with a sufficiently large separation. In fact, only 29 out of 73 galaxies have Einstein radii larger than the observed image separation in B0850+054.

3.7 Parameters Constraints

Our second approach to investigating the central image of B0850+054 is to set constraints on the values of r_b and γ for the lensing galaxy by exploring the parameter space. By assuming a Nuker law for the mass profile of the lensing galaxy, we take a set of initial values in the 6-parameter space $(\alpha, \beta, \gamma, r_b, \kappa_b, \sigma)$ and optimize for the source position s and convergence κ_b. The fitted models aim to reproduce the observed separation and flux ratio between the two bright components A and B, and the mass distribution of the lensing galaxy can be obtained. Since the e-MERLIN observation provides sensitivity that is an order-of-magnitude better, the results are able to set a tighter constraint on the mass distribution of the lensing galaxy in B0850+054.

The values of the parameters are chosen based on previous studies (e.g. Keeton 2003; Quinn et al. 2016) and how well they agree with M87, which is able to reproduce the observed quantities in Section 3.6. Figure 3.6 shows the distributions of parameter values in the Keeton (2003) sample and the parameter values of M87. We have taken the outer power-law index as isothermal

(β = 1) and the sharpness of break at α = 2. The modelling is also performed with three different scenarios in the central region of the galaxy – without a black hole, and with black holes of velocity dispersions σ = 100 and 200 km s^{-1}. By inserting a black hole into the mass distribution, the central image is highly demagnified due to the very steep region towards the centre of the lens. The selected velocity dispersions correspond to black hole masses of 8.92×10^6 $M_\odot$ and 1.20×10^8 $M_\odot$ respectively, using the Gebhardt relation for black hole masses and velocity dispersions (Equation 1.22). After fitting for the source position s and the convergence κ_b with different black hole masses, we release the third parameter to fit for the sharpness of break α. We also try to use a different initial value for the sharpness of break at α = 3, and repeat the same procedures to explore the parameter space more thoroughly.

By varying the logarithm of the break radius r_b from -3 to 0 at a step of 0.05 and varying the inner power-law index γ from 0 to 1 at a step of 0.025, modelling results are shown in Figure 3.7 and 3.8. Blank regions in the graphs indicate that models have failed ($\chi^2 > 5$), and a lot of models have been excluded because of this. The colours in the plots show the logarithm of the A:C flux ratio at different r_b and γ values. The red contour lines represent the detection limit of the central image in the e-MERLIN 5-GHz observation. The results are similar for the initial conditions α = 2 and α = 3. Mass distributions without a black hole and and with a black hole of velocity dispersion σ = 100 km s^{-1} are able to reproduce the observed source separation and flux ratio at certain ranges of r_b and γ. As the break radius

increases, the inner power-law index also increases to reproduce the same flux ratio, indicating that the inner region of the mass distribution becomes closer to isothermal. The insertion of a black hole of velocity dispersion σ = 100 km s^{-1} wipes out the region where γ is large, heavily demagnifying the central image, making it impossible to be detected. A black hole of σ = 200 km s^{-1} totally wipes out the whole parameter space, so that the central image can never be detected with the observed separation and A:B flux ratio as constraints. Attempting to fit for the sharpness of break α results in certain regions in the parameter space to be successfully fitted and reproduce the data constraints. Nevertheless, these regions are only results from badly fitted data with negative values of α, so they are ignored. The non-detection of the central image in the e-MERLIN 5-GHz observation at A:C ratio $\sim$ 2250/1 indicates that the either the central image is heavily demagnified by a central SMBH in the lensing galaxy, or that the galaxy has a near-isothermal central region and a small break radius. Since the true values of r_b and γ are not known, there are not enough constraints to obtain the true black hole mass in the centre of the galaxy.

3.8 Conclusion

We have closely studied the gravitational lens system B0850+054 to try to detect the central image and perform mass modelling. The central image is not detected in the e-MERLIN 5-GHz map where the rms noise level is

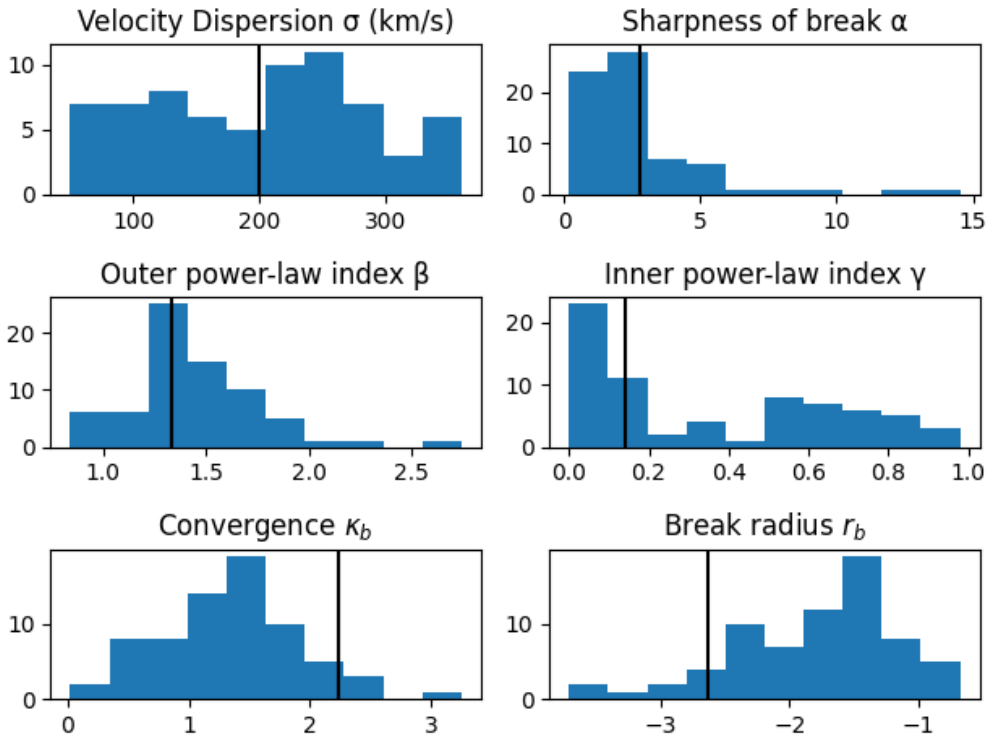


Figure 3.6: Distributions of Nuker-law parameters of early-type galaxies in the Keeton (2003) sample. The black vertical line of each graph shows each parameter value of M87, which is successful in reproducing the observed source position and flux ratio.

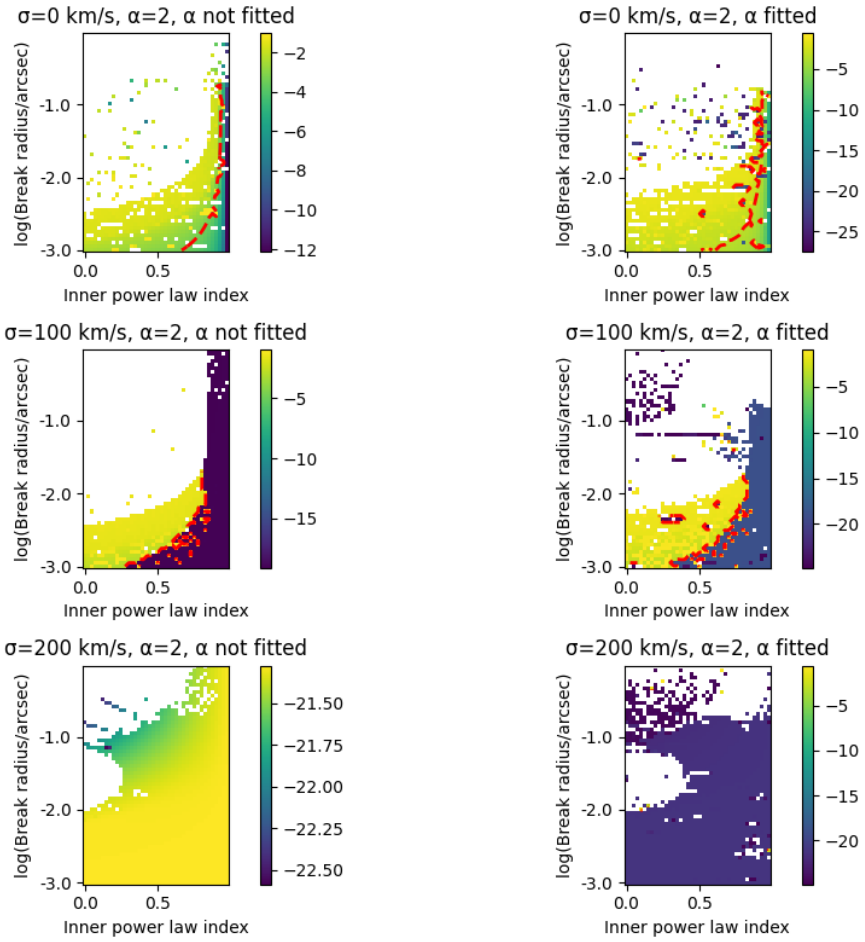


Figure 3.7: Mass modelling results by varying r_b and γ to optimize for the convergence κ_b and source separation s. Each graph shows the logarithm of the break radius r_b plotted against the inner power-law index γ. The colours in the plots show the logarithm of the reproduced image flux ratio $\log(\mu_C/\mu_A)$ and the red contour represents the detection limit of the e-MERLIN 5-GHz observation of B0850+054. The models in each plot are constrained using the observed source separation and A:B flux ratio. Initial values of the modelling are compromised between the Keeton (2003) sample and M87 Nuker-law parameters that have successfully reproduced the data constraints. Different black hole velocity dispersions are used in modelling attempts with α = 2, and fitting of α is also attempted. The varied initial conditions are listed in the title of each plot.

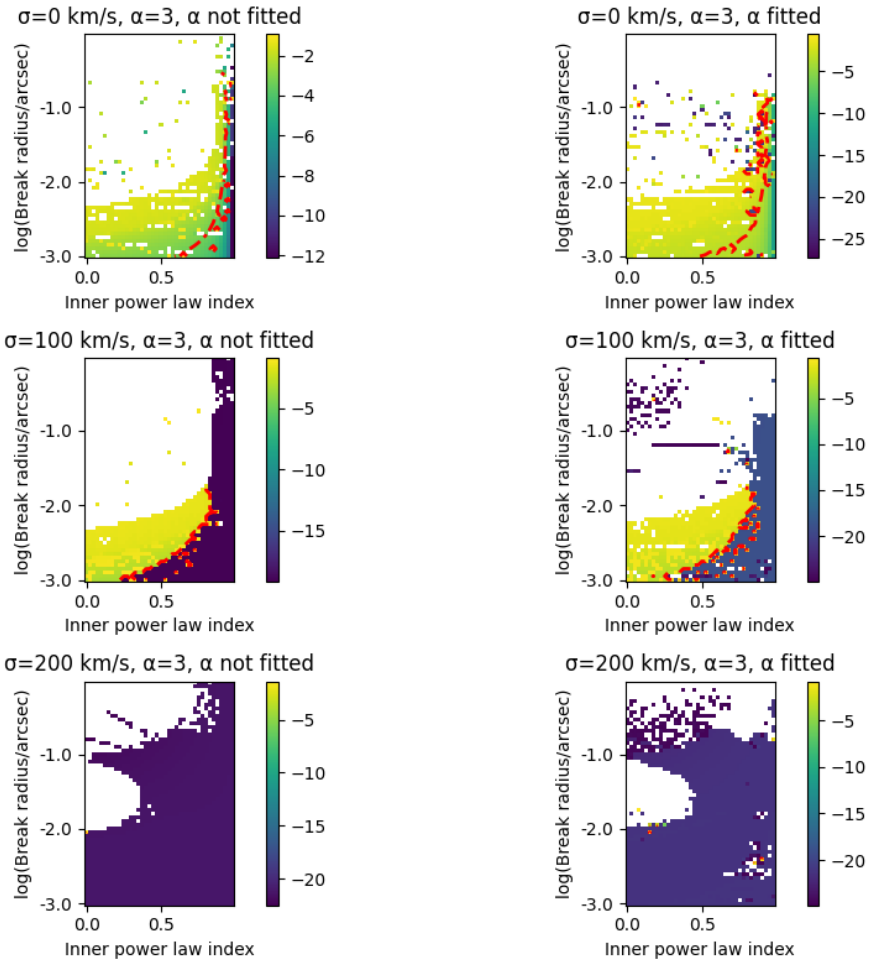


Figure 3.8: Similar plots to Figure 3.7 but with $\alpha = 3$. Again, each graph shows the logarithm of the break radius r_b plotted against the inner power-law index γ. The colours represent $\log(\mu_C/\mu_A)$ and the red contour represents the A:C flux ratio constraint that the e-MERLIN 5-GHz observation provided. Results are similar to Figure 3.7, indicating that either the the galaxy has a close-to-isothermal inner region with a small break radius, or a central SMBH exists at the core of the galaxy to heavily demagnify the central image.

6.76 μJy beam^{-1}. Nevertheless, the new e-MERLIN radio map helps set tighter constraints on the inner region of the lensing galaxy. We first investigate the Keeton (2003) sample for early-type galaxies to model the lens system, and M87 is the only galaxy that is able to reproduce the observed source separation and flux ratio. The Nuker-law parameters of M87 are then compared to the Keeton (2003) sample to find compromised initial conditions for the next modelling procedure. By fixing (σ, α, β) and varying (r_b,γ), we optimize for the convergence κ_b and the source separation s to reproduce the observed source separation and flux ratio. With the fact that the central image is not detected in the e-MERLIN 5-GHz map, results indicate that the galaxy either contains a central SMBH that heavily demagnifies the central image, or that the galaxy has a near-isothermal inner region and a small break radius.

Chapter 4

Radio Observation of HE 0435-1223

4.1 Introduction

The four-image gravitational lens system HE 0435-1223 was selected in the Hamburg/ESO digital objective prism survey to identify high-probability QSO candidates (Wisotzki et al. 2000). High resolution images at optical wavelengths were taken with the 6.5-m Baade Telescope and the lens system was confirmed to contain four QSO components at redshift z = 1.69. Morgan et al. (2005) obtained the lens redshift z = 0.45 and found that the lensing galaxy inhabits a group of 10 galaxies. Figure 4.1 shows the VLA radio map of HE 0435-1223, where the four components are arranged in a cross-shape configuration and the maximum separation between components is

2.54 arcsec (Jackson et al. 2015). Using a single-galaxy deflector model, Sluse et al. (2012) reproduced the positions of the lensed components successfully.

Similar to radio emissions from galaxies, far-IR emissions have larger characteristic sizes than their optical/mid-IR counterparts, as emissions come from regions of lower temperatures. Therefore, both radio and far-IR sources are not subject to microlensing as their characteristic sizes are much larger than the characteristics sizes of microlensing caustics. Fadely & Keeton (2012) obtained the flux ratios between components in the far-IR to study the lens system for this reason. Using the flux ratios as constraints, lens modelling was carried out using a softened power-law for the central lensing galaxy, together with a singular isothermal model for a nearby galaxy G22 and a shear component for the shear field of the galaxy cluster. However, the observed flux ratios between the lensed images were not successfully reproduced. A different result for the observed flux ratios between components was obtained using the VLA at 5 GHz (Jackson et al. 2015). Using an ellipsoidal power-law with an external shear for the central lensing galaxy and a singular isothermal sphere for G22, modelling was carried out with a point-like source. Nevertheless, the model failed to reproduce the observed flux ratios and the residual was significant. Jackson et al. (2015) suggested that the flux anomaly in HE 0435-1223 can be explained by invoking substructure using a pseudo-Jaffe profile over a wide range of masses; or alternatively, by increasing the source size as the size of the caustic is less than ~ 400 mas. An optimal FWHM of 80 ± 5 mas and an upper limit of 200 mas for the source

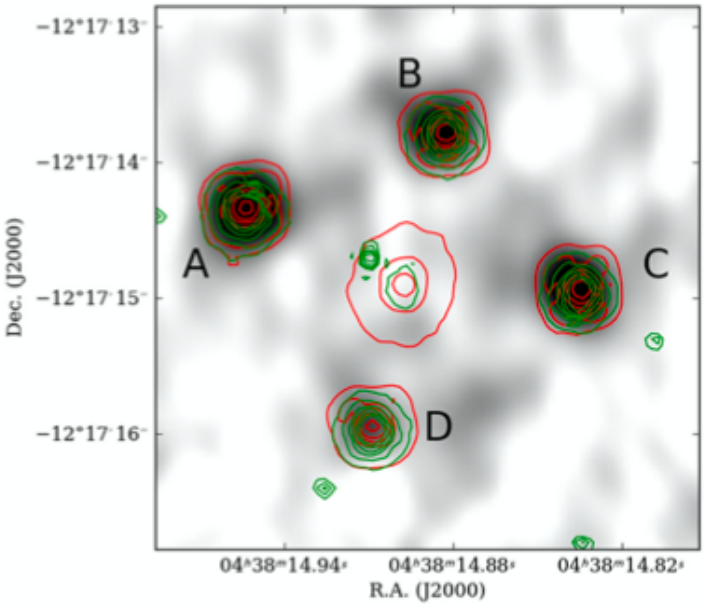


Figure 4.1: VLA grey-scale image of HE 0435-1223 at 5 GHz. Archival HST Near-Infrared Camera and Multi-Object Spectrometer (NICMOS; red contours) and Advanced Camera for Surveys (ACS; green contours) are plotted on top of the radio map. The radio map shows the four components of HE 0435-1223 in a cross-shaped configuration. Reproduced from Jackson et al. (2015).

size were obtained after considering the non-Gaussianity and correlations of noise in the proximity of the lensed images.

In this chapter, we present new e-MERLIN data on HE 0435-1223 to investigate the source size. As radio interferometers with longer baselines are more insensitive to large-scale structures, we can obtain the lower limit on the source size if the source is resolved out in the radio map. Consequently, we can determine the upper limit on the brightness temperature of the source using the rms noise level of the map. If the source is detected in the radio map, we can obtain an upper limit on the source size, and a lower limit on its brightness temperature. A low brightness temperature ($< 10^4$ K) indicates that the source is powered by star formation while a high brightness temperature ($\sim 10^6 - 10^9$ K) indicates an AGN mechanism at work.

4.2 Observations

A total of 11 observations on HE 0435-1223 were done from October 2020 to January 2021, using e-MERLIN at L-band (1.5 GHz). The L-band observation has a frequency range of $1.25 - 1.77$ GHz. Similar to the e-MERLIN C-band, the maximum bandwidth is 512 MHz with an average integration time of 1 s. As observations with the L-band are done at a lower frequency, bandwidth smearing effects are more serious. The bandwidth is therefore divided into 8 SPWs. Dividing each SPW into 128 channels and 4 polarizations gives a channel width of 125 kHz. Observing with the L-band generally

results in a lower angular resolution than observing in the C-band. In addition, there is a significant amount of RFI that affects observations in the L-band, e.g. radar transmissions, satellites.

The total on-source integration time was ~ 57 hours and the u-v coverage on the source is plotted in Figure 4.2. Since the Lovell Telescope was used in only two out of all the observations (Table 4.1) and observations were seriously affected by RFI, extra observation time was given. Observations were done on the target source (HE 0435-1223) and a nearby phase-reference source (J0434-1442) alternately on a cycle of ~ 7 minutes and ~ 3 minutes respectively. A bandpass calibrator (J1407+2827) was observed for the bandpass correction. The flux calibrator (3C286) was observed, but not used to determine the flux density scale as its model was not found during calibration process. The overall flux density scale is thus determined using the same method in the previous chapter – by measuring the flux and spectral index of the bandpass calibrator (J1407+2827), and then confirmed with its absolute flux scale (e.g. Langston et al. 1990; Helmboldt et al. 2007) to match with the flux density scale. As the target source is radio-quiet and the datasets consist of mostly noise, least-square fitting may fail when trying to find solutions for such data during self-calibration. Therefore, self-calibration was done on the phase-reference source instead. Self-calibration on the nearby phase-reference source is required if the phase-reference source is not exactly a point source. This procedure improves the quality of the map by finding the phase solutions for the phase-reference source and interpolating them to

the target source map. Moreover, the nearby phase-reference source has a greater intrinsic brightness so that self-calibration is less likely to fail. After the solutions to the phase and amplitude corruptions in an observation are found, they can be applied to the target source to produce the final map.

4.3 Calibration

Initial calibration was done using the e-MERLIN CASA pipeline (Moldon 2021). The calibrated data were loaded into AIPS using the task FITLD. After careful inspections on the scan lists and the calibrated u-v data using the tasks LISTR and SPFLG respectively, roughly 10-15% of the data were flagged out automatically in each observation. However, a lot of remaining bad data existed as the pipeline cannot filter out all the RFI. The tasks SPFLG and POSSM were used to closely eyeball the u-v data in each baseline. Bulk bad data across the whole spectral window were flagged out using the task UVFLG and remaining bad data were flagged out using the interactive TV display in SPFLG. After flagging, an extra 15% data were flagged out, therefore quite significantly affecting the rms noise level of some of the observations (Table 4.1). The amount of bad data was not only affected by the presence of RFI in the L-band, but also the low elevation of the source, as the antennas were likely to detect more artificial signals when they were pointed at low elevations. The low elevation of the source also affected the beam shape, which will be further discussed in Section 4.5.

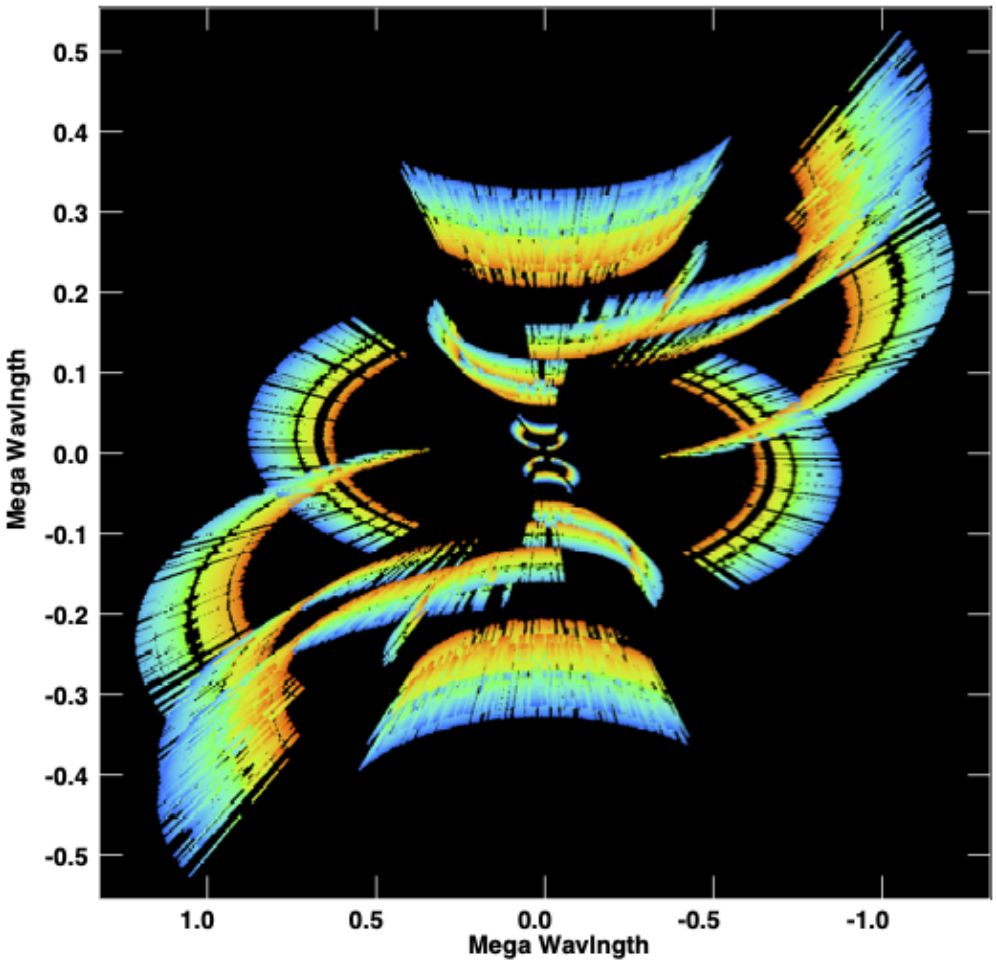


Figure 4.2: The u-v coverage of the observations on HE 0435-1223. Units of the u-v plane are in mega wavelengths, which correspond to 200 km in e-MERLIN 1.5-GHz observations. Again, the colours represent different IFs and channels, with red indicating the lowest frequency and blue indicating the highest frequency.

The determination of the flux scale followed the same procedures in Section 3.3, due to the lack of flux density model for the flux calibrator (3C286) in AIPS. The target source (HE 0435-1223) has a flat spectrum at 1.5 GHz, so the bandpass calibrator (J1407+2827) was used to create the flux density scale. The fluxes of J1407+2827 at different frequencies were measured with POSSM and its spectral index was calculated using Equation 3.1. The task SETJY was used to set the flux density scale after confirming that the observed fluxes match with its absolute flux density scale in previous surveys (e.g. Langston et al. 1990; Helmboldt et al. 2007). The task BPASS was also used to calibrate the sensitivities within the bandpasses as a function of frequency, and applying them to correct for errors.

4.4 Imaging

As the target source (HE 0435-1223) is radio-quiet, direct self-calibration on the source may result in a failure of `CLEAN`ing. An iterative cycle of `CLEAN` and self-calibration on the phase calibrator (J0434-1442) was used to find the phase and amplitude corrections for each observation. The corrections were then applied to the target source to obtain a final map. The task IMAGR was first used on the calibrated dataset to map J0434-1442. The cell size of the map was set to be 0.05 arcsec × 0.05 arcsec, and the size of the image was set to be 512 pixels × 512 pixels. The number of iterations in each `CLEAN` cycle was set to be 5000. By obtaining the number of clean componenets up

to the first negative value with the task PRTCC, the task CALIB was used on J0434-1442 to find solutions by self-calibration. A long solution interval (1 hour) was initally used in phase self-calibration, and is gradually decreased to 10 minutes. A long solution interval of 1 hour hardly changes the map, as the phase corruptions were at much shorter timescales than the solution interval. At solution interval of 10 minutes, the true phase solutions could be found as the interval is shorter than the phase coherence time. It is also important to ensure that the number of free parameters does not exceed the number of constraints by avoid using a solution interval that is too short. After using CALIB, the task CLCAL was used to interpolate the solutions to the data in a calibration table, where an empty table was created using the task INDXR. INDXR was used because the interval of the calibration table could be set to be 0.1 minute. This avoided the loss of solutions during the interpolation to the calibration table. The output calibration table was used in the next `CLEAN` cycle to map the phase calibrator.

After the final cycle of phase self-calibration, amplitude self-calibration with a solution interval of 15 minutes was carried out on J0434-1442 to correct for residual errors. We kept the solution interval of the amplitude self-calibration to be larger than the final solution interval of phase self-calibration as the amplitude corruption was less severe, and to ensure that the number of constraints was greater than that of the free parameters. With the calibration table that contained the final phase and amplitude solutions, the final cycle of `CLEAN`ing was carried out on the target source (HE 0435-

1223), to produce the final map. The rms noise levels of the maps in each observation are listed in Table 4.1. The rms noise levels of some observations are exceptionally large, mainly due to the lack of the Lovell telescope in these observations, and the substantial amount of flagged data. When combining the u-v data to produce the final map, we excluded 4 observations, where their rms noise levels are greater than 50 μJy. Table 4.2 shows the beam sizes and the position angles of the remaining observations. The data of these observations were combined using the tasks SPLIT and DBCON. SPLIT was used on all of the datasets to reduce the sizes of the datasets by converting a multi-source file in to a single source dataset. The task ignored all the flagged data and applied the calibrations to the chosen target source. DBCON was then used to concatenate two u-v datasets together. Only two datasets could be concatenated together at one time, so DBCON was used repeatedly until all the desired datasets were combined. Figure 4.3 shows the final e-MERLIN grey-scale radio map of HE 0435-1223 at 1.5 GHz and the noise level of the map is 11.63 μJy beam^{-1}. The estimated rms noise level of an e-MERLIN observation (no Lovell Telescope) at L-band with 57 hours on the source is $\sim$ 9.2 μJy beam^{-1}. The worse sensitivity in the actual observation can be explained by the poor qualities of the observations resulting in a large amount of flagged data ($\sim$ 25 – 30%) and residual noise that exist in the datasets even after calibration. A robust weighting (Briggs 1995) of zero robustness was used in `CLEAN`ing to compromise between natural and uniform weighting (Section 2.4.2). To produce the final map of HE 0435-1223, a restoring beam

of a 480×106 mas^2 and a position angle of 14.46° was chosen by taking the average of the `CLEAN` beams of the observations.

4.5 Non-detection

The VLA radio map of HE 0435-1223 at 5 GHz detects the four-image lens system at a noise level of 3 μJy beam^{-1} with a 6-hour on-source time (Figure 4.1; Jackson et al. 2015). The beam is circular and of FWHM 0.45 arcsec, indicating that the emission region of the background source is at scale smaller than the beam size. However, when the source is observed using e-MERLIN at 1.5 GHz, it is not detected. The rms noise level is 11.63 μJy beam^{-1}, so we can determine the 3-σ detection limit of HE 0435-1223 at 34.89 μJy beam^{-1}. The beam is elliptical with major and minor axes of 480 mas and 106 mas respectively (Figure 4.3). Since the large-scale structure of a source is not recorded when the apparent source size is larger than the resolution of the observation, the non-detection of HE 0435-1223 indicates that the angular size of HE 0435-1223 at 1.5 GHz is bigger than the beam size. As the source size tells us where the emission originates, results indicate that radio emission of HE 0435-1223 comes from an extended region around the central SMBH, and this supports the possibility that in HE 0435-1223, emission is primarily due to star formation. The e-MERLIN 1.5-GHz observation on HE 0435-1223 shows an improvement in resolution by a factor of 3 compared to the VLA 5-GHz observation. Although the observations were done at low elevation re-

Obs. Date	Lovell Telescope	Included?	rms (μJy beam^{-1})
Oct. 03	✓	✓	14.36
Oct. 19	×	✓	35.58
Dec. 23	×	×	50.46
Dec.24	×	×	77.81
Dec. 25	×	×	58.20
Dec. 27	×	×	87.86
Jan. 01	×	✓	37.98
Jan. 02	×	✓	40.37
Jan. 03	×	✓	38.48
Jan. 11	✓	✓	36.44
Jan. 25	×	✓	36.65

Table 4.1: Information of the e-MERLIN 1.5-GHz observations on HE 0435-1223. The Lovell Telescope was only included in two of the observations. The rms noise level are also listed. Four observations are excluded when producing the final map of HE 0435-1223 due to their poor sensitivities (> 50 μJy beam^{-1}).

Obs. Date	bmaj (arcsec)	bmin (arcsec)	bpa (deg)
Oct. 03	0.413	0.117	18.70
Oct. 19	0.398	0.114	11.87
Jan. 01	0.66	0.080	14.79
Jan. 02	0.418	0.105	12.89
Jan. 03	0.428	0.102	12.89
Jan. 11	0.630	0.120	14.65
Jan. 25	0.416	0.104	15.41
Average	0.480	0.106	14.46

Table 4.2: Beam sizes and position angles of the selected HE 0435-1223 observations. The observation dates, beam major axes, beam minor axes and beam position angles are listed from left to right. The averages of these values are the size and position angle of the restoring beam used to produce the final map.

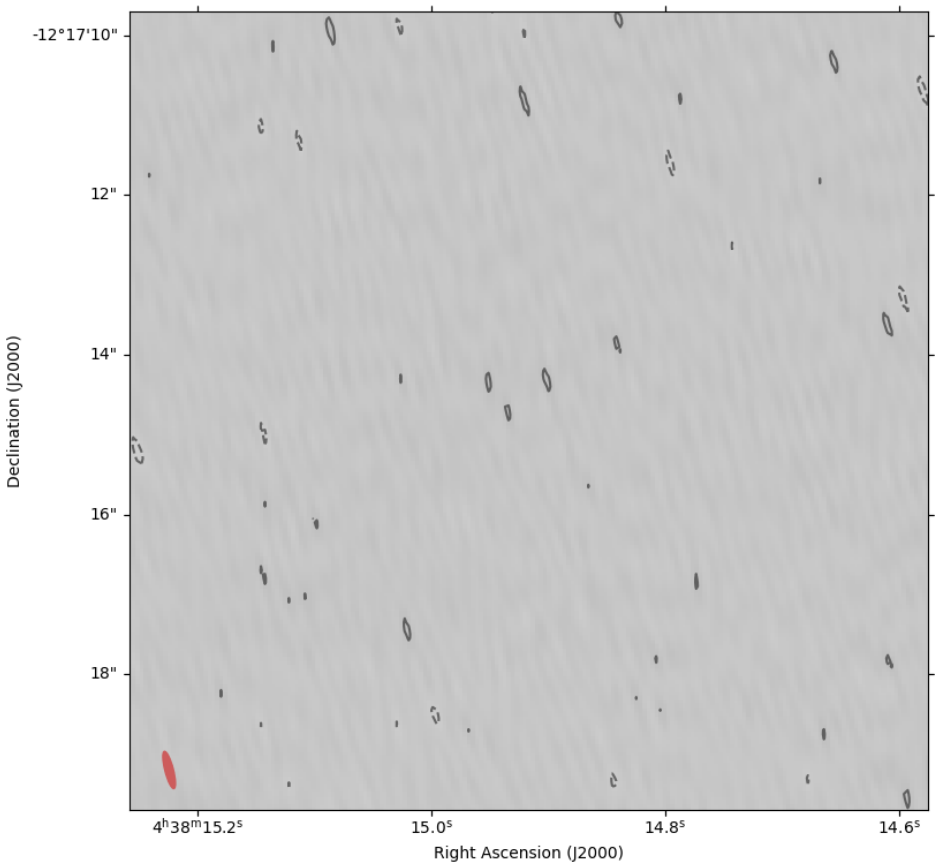


Figure 4.3: e-MERLIN 1.5-GHz grey-scale image of HE 0435-1223 with a restoring beam of 480×106 mas^2 at a position angle of 14.46°. The data are weighted with zero robustness (Briggs 1995) and the rms noise level is 11.63 μJy beam^{-1}. The restoring beam is plotted in red at the bottom left corner. Contours are plotted at multiples (1,1,2,4,8,16,32,64,128) of 3σ where σ is the rms noise level in the map.

sulting in a worse resolution and an elliptical beam, the non-detection of the source allows us to determine the size of the emission region in HE 0435-1223 and its possible emission mechanism. Wucknitz & Volino (2008) had similar results when observing the lens system J1131-1231 where the background source was observed in MERLIN observations but showed non-detection in EVN e-VLBI observations. They suggested that the absence of AGN in the lensed components indicates that the background source is powered by star formation.

Hartley et al. (2019) detected jet activities within a strongly lensed RQQ HS 0810+2554, which set an upper limit on the source size. Using Equation 1.26, a lower limit on the brightness temperature at 8.4×10^6 K was obtained. As AGN mechanisms give a brightness temperature of $\sim 10^6 - 10^9$ K (Hartley et al. 2019), the brightness temperature of HS 0810+2554 indicates that AGN mechanism is at work for that source. Falcke et al. (1996), Ulvestad et al. (2005), and Chen et al. (2023) also measured the brightness temperatures of radio-quiet quasars to determine the emission mechanisms at work for different sources.

The non-detection of the source at $z = 1.69$ in the e-MERLIN 1.5-GHz observation gives us a detection limit on the source at 34.89 μJy beam^{-1}, and a lower limit on the beam size 408×106 mas^2. Using Equation 1.26, an upper limit on the brightness temperature at 1.74×10^3 K is obtained. The low brightness temperature ($< 10^4$ K) indicates that the emission mechanism is likely due to star formation, as opposed to AGN mechanism, magnetically

heated corona or other emission mechanisms.

4.6 Conclusion

We have observed the strongly lensed system HE 0435-1223 with e-MERLIN at 1.5 GHz to investigate its emission mechanism. The source is not detected in the map where the rms noise level is 11.63 μJy beam^{-1}, with a beam size 408×106 mas^2. The non-detection of the source indicates that the source size is at a characteristic scale larger than the beam size, thus the emission comes from an extended region around the central SMBH. The detection limit of the observation at 34.89 μJy beam^{-1} sets a limit on the brightness of the source, and a lower limit on the source size can also be obtained at 408×106 mas^2. These help us obtain an upper limit on the brightness temperature at 1.74×10^3 K using Equation 1.26. The low brightness temperature ($< 10^4$ K) therefore indicates that the emission in HE 0435-1223 is due to star formation.

Chapter 5

Summary and Future Work

In this thesis, two gravitationally-lensed quasars – CLASS B0850+054 and HE 0435-1223 are observed with e-MERLIN. Quasars are distant point-sources, and are extraordinary bright. They can greatly facilitate the study of gravitational lensing, as lensed quasars are easily recognised. Using radio interferometry to observe these sources provides incredible resolution with the length of the baselines an interferometer can have.

5.1 CLASS B0850+054

We have presented e-MERLIN 5-GHz data on CLASS B0850+054 in search of the central image of the lens system and have performed mass modelling to study the inner mass profile of the lensing galaxy. Burke (1981) suggested that when the central mass distribution is less steep than $\rho \propto r^{-2}$, there is

an additional central image in the vicinity of the lensing galaxy. However, this image is often highly demagnified and is very difficult to detect. CLASS B0850+054 is a two-image lens system with a flux ratio $\sim 6.15:1$ between the two bright components. Such a system is a suitable candidate for central image detection as the central image forms further away from the lens centre. Although the e-MERLIN 5-GHz observation shows a non-detection for the central image, it has helped set tighter constraints on the detection limit of the central image. By investigating the Keeton (2003) sample for early-type galaxies, we have obtained a set of Nuker-law parameters that is able to reproduce the observed source separation and flux ratio. The parameters are compared to the distributions of parameters in the Keeton (2003) sample to determine initial parameter values for the next step of modelling. By varying the break radius r_b and inner power-law index γ at different black hole masses $\mathrm{M_{BH}}$, we optimize for the source separation s and convergence κ_b to try to reproduce the observed source separation and flux ratio. Results show that the non-detection of the central image in the observation indicates that the lensing galaxy contains a SMBH that heavily demagnifies the central image, or the central region of the lensing galaxy is small and near-isothermal. However, the black hole mass, and the values of r_b and γ cannot be determined due to the insufficient constraints that the e-MERLIN 5-GHz observation has. To find these values, an observation with even better sensitivity is needed to actually detect the central image, allowing us to set tighter constraints on the inner mass profile of the lensing galaxy.

5.2 HE 0435-1223

HE 0435-1223 was observed with e-MERLIN at 1.5 GHz to set a limit on the brightness temperature of the background source. By observing the lensed quasar with radio interferometry at different resolutions, we can obtain the lower limit of the source size as the source becomes undetected at a certain resolution. The lower limit of the source size allows us to set an upper limit on the brightness temperature of the background source, which tells us the possible emisison mechanism at work. Hartley et al. (2019) obtained a brightness temperature of at least 8.4×10^6 K for the lensed radio-quiet quasar HS 0810+2554 and detected jet activity from the source.

The data of HE 0435-1223 shows a non-detection of the source, indicating that the background source is larger than the beam size 480×106 mas^2. We have thus set a detection limit on the background source at 34.89 μJy beam^{-1}. An upper limit on the brightness temperature 1.74×10^3 K is obtained. This tells us that the emission in HE 0435-1223 is most likely due to star formation around the central SMBH. There is available VLBI data on HE 0435-1223, in which a non-detection is also expected as the resolution of the observation is even better. The data can be calibrated to confirm that the source is not detected but cannot set a tighter constraint on its brightness temperature.

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